

A PROJECT REPORT ON

AI-Powered CV and Job Description Matcher

An AI-Based Resume Analysis and Job Screening Assistance System

Submitted in partial fulfillment of the requirements for the award of the degree of

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CERTIFICATE

This is to certify that the project report entitled "**AI-Powered CV and Job Description Matcher: An AI-Based Resume Analysis and Job Screening Assistance System**" submitted by **[Student Name]** (Roll Number: **[Roll Number]**) in partial fulfillment of the requirements for the award of the degree of **[Course/Degree]** in **[Department]** at **[College/University]** is an authentic record of the academic software engineering work carried out under my supervision and guidance during the academic year **[Academic Year]**.

The results embodied in this report have been verified for technical soundness, software engineering rigor, and ethical compliance, and have not been submitted elsewhere for the award of any other degree or diploma.

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CANDIDATE DECLARATION

I hereby declare that the software project entitled "**AI-Powered CV and Job Description Matcher: An AI-Based Resume Analysis and Job Screening Assistance System**" presented in this report is an original work developed by me under the academic guidance of **[Project Guide]**, Department of **[Department]**, **[College/University]**.

I explicitly confirm that:

1. The conceptual design, system architecture, database models, algorithmic scoring mechanisms, ATS audit checks, and user interface mockups documented herein represent bona fide academic investigation and technical implementation.
2. All literature, libraries, algorithms, third-party software frameworks, and application programming interfaces (APIs) consulted or incorporated during this project have been fully acknowledged and cited according to standard academic conventions.
3. The analytical scoring models, match percentages, and estimated shortlist probabilities generated by this system are diagnostic estimations designed for candidate feedback and do not constitute absolute guarantees of corporate employment or interview selection.
4. This work has not formed the basis for the award of any other academic degree, diploma, associate-ship, or fellowship at this or any other university.

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ABSTRACT

In the contemporary global recruitment ecosystem, the volume of digital job applications submitted to corporate requisitions has grown exponentially, regularly exceeding several hundred resumes per open position. To cope with this immense influx, enterprise organizations heavily deploy Applicant Tracking Systems (ATS) to automate initial candidate screening, filtering, and ranking. However, this reliance on automated parsing introduces substantial friction and systemic information asymmetry. Qualified candidates frequently fail initial automated screenings due to lexical mismatches (e.g., using synonymous terminology or unexpanded acronyms), unfavorable document formatting (such as complex tables, multi-column layouts, or unreadable headers), or subtle omissions of required technical competencies. Conversely, manual CV-JD cross-referencing is highly tedious, subjective, and prone to oversight.

To resolve these challenges, this project presents the design, architectural formulation, and implementation of the **AI-Powered CV and Job Description Matcher**, an intelligent, web-based software system that combines natural language processing (NLP), semantic knowledge graphs, heuristic rule engines, and Large Language Model (LLM) abstractions to perform rigorous, explainable resume-job description compatibility analysis. The system accepts candidate CVs across diverse file formats (PDF, DOCX, TXT) alongside target job requisitions, extracts unstructured textual content, detects potential ATS parsing hazards, and normalizes candidate qualifications against requisition demands.

The cognitive core of the proposed system introduces a multi-tier matching and scoring architecture. Rather than relying on naive keyword counting, the engine differentiates between **Exact Matches**, **Canonical Synonyms** (e.g., *Postgres* ↔ *PostgreSQL*), and **Partial Sibling Technologies** (e.g., correlating *RabbitMQ* experience against a *Kafka* requirement via taxonomy graphs), while explicitly flagging missing mandatory and preferred skills. Furthermore, the system incorporates a 12-point ATS structural compliance audit, computes a transparent 7-factor weighted match score (0–100%), models an estimated shortlist screening probability, and synthesizes actionable, quantified bullet-point improvement suggestions equipped with strict anti-hallucination guardrails.

The software is engineered using a robust full-stack architecture comprising a React 18 and TypeScript single-page frontend styled with Tailwind CSS, a high-performance Python 3.11 and FastAPI backend, and an extensible persistence layer built on SQLAlchemy supporting SQLite and PostgreSQL. The application empowers job seekers to identify critical competency gaps, optimize document ATS compliance, and articulate their genuine qualifications with maximum clarity prior to official submission.

Keywords: Resume Analysis, Job Description Matching, Applicant Tracking Systems (ATS), Natural Language Processing (NLP), Semantic Similarity, Large Language Models (LLM), Keyword Extraction, Recruitment Technology, Decision Support System.

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1. BACKGROUND OF THE STUDY

1.1 Introduction

The contemporary employment and talent acquisition landscape has experienced a profound structural transformation driven by the ubiquity of digital job boards, professional networking platforms (e.g., LinkedIn, Indeed, Glassdoor), and one-click application portals. While these digital platforms have substantially democratized access to career opportunities, they have simultaneously catalyzed an unprecedented surge in application volumes. Enterprise organizations and growth-stage technology companies routinely receive hundreds—and in many high-visibility corporate postings, thousands—of curriculum vitae (CVs) and resumes for every single open requisition.

Faced with this immense administrative bottleneck, human talent acquisition teams are physically unable to perform exhaustive, manual line-by-line evaluations of every submitted dossier. Consequently, enterprise recruitment workflows have become heavily reliant on automated **Applicant Tracking Systems (ATS)**. These automated software suites act as initial algorithmic gatekeepers, ingesting digital documents, extracting candidate information, filtering out non-conforming profiles, and ranking candidates based on keyword frequency, lexical overlap, and basic educational thresholds.

However, legacy ATS architectures and unassisted manual screening processes suffer from critical systemic flaws. Traditional keyword matching relies on exact string equality or basic stemming, which frequently penalizes candidates who express equivalent competencies using synonymous phrasing, alternative industry terminology, or unexpanded acronyms. Furthermore, the recruitment screening phase often emphasizes surface-level keyword presence while failing to evaluate the deeper semantic context of a candidate's demonstrated project impact, engineering responsibilities, and seniority trajectory. This dynamic creates a profound information asymmetry and communication breakdown between qualified job seekers and prospective employers.

1.2 Background

The evaluation of a candidate's resume against a complex Job Description (JD) represents a non-trivial information retrieval and semantic matching challenge. A typical technical job description encapsulates multiple multidimensional constraints, including core programming languages, modern architectural frameworks, cloud infrastructure platforms, data storage engines, minimum years of progressive industry experience, educational credentials, domain-specific regulatory knowledge, and essential collaborative soft skills.

A prevalent issue in recruitment science is the *competency articulation gap*: a candidate may genuinely possess the requisite technical expertise, problem-solving acumen, and hands-on domain experience to excel in a role, yet fail to pass the initial automated or human screening. This failure commonly occurs because the candidate's resume lacks the precise lexical keywords prioritized by an automated filter, presents critical accomplishments in passive or non-quantified phrasing, or embeds crucial skills in visual formatting structures (such as multi-column tables, text boxes, or graphical header blocks) that corrupt ATS document parsers. Without intelligent, pre-submission diagnostic tooling, job seekers are forced to navigate the hiring pipeline blindly, resulting in high application attrition, demoralization, and substantial loss of qualified talent for hiring organizations.

1.3 Existing Recruitment Process

The standard end-to-end recruitment lifecycle across modern corporate and technology enterprises follows a structured, multi-stage pipeline comprising the following eight distinct operational phases:

- **Stage 1: Requisition Creation & Job Posting:** Hiring managers collaborate with talent acquisition specialists to formulate the official Job Description, defining core responsibilities, mandatory technical requirements, preferred qualifications, and organizational expectations.

- **Stage 2: Candidate Application:** Job seekers discover the requisition via corporate careers portals or external job boards and submit their profile data alongside digital CV/resume documents (PDF or DOCX).
- **Stage 3: Resume Submission & Ingestion:** The corporate ATS ingests the submitted document, parsing raw binary streams into plain text and mapping extracted blocks into database fields.
- **Stage 4: Automated ATS Screening & Parsing:** The ATS executes rule-based Boolean queries, keyword density filters, and credential checks, assigning preliminary candidate rankings and automatically rejecting sub-threshold submissions.
- **Stage 5: Recruiter Manual Review:** Human recruiters rapidly scan shortlisted resumes—allocating an industry-average of 6 to 7 seconds per CV—to verify high-level role alignment, current employer pedigree, and keyword prominence.
- **Stage 6: Technical / Hiring Manager Screening:** Resumes that pass the initial recruiter filter are forwarded to the hiring manager and lead engineers for in-depth evaluation of project relevance and architectural depth.
- **Stage 7: Multi-Round Structured Interviews:** Candidates undergo sequential evaluations, including technical coding assessments, system design interviews, behavioral reviews, and cultural alignment discussions.
- **Stage 8: Offer & Final Selection:** Background checks, reference verifications, and compensation negotiations culminate in the final employment contract.

1.4 Role of ATS in Recruitment

Applicant Tracking Systems function as the backbone of modern corporate hiring infrastructure. At their core, ATS platforms execute four fundamental operations: document text parsing, entity extraction, rule-based keyword matching, and candidate ranking. When an applicant uploads a CV, the ATS attempts to deconstruct the unstructured document into structured semantic entities: Contact Information, Professional Summary, Work History, Education, and Technical Skills.

Critical Limitations of Commercial ATS Engines:

Despite their widespread adoption, commercial ATS engines exhibit notable technical limitations. Most legacy parsers rely on inflexible optical or layout assumptions. Complex typographic elements—such as multi-column layouts, floating text frames, nested tables, non-standard section headers (e.g., 'Core Competencies' vs 'Technical Skills'), and graphical icons—frequently cause text extraction routines to scramble reading orders, concatenate disparate columns, or completely discard entire sections. Furthermore, ATS keyword matching algorithms rarely possess deep semantic awareness; they fail to recognize that a developer with extensive *PostgreSQL* expertise is inherently qualified for a role specifying *Postgres*, or that extensive experience in *RabbitMQ* distributed queuing provides strong foundational competence for a role seeking *Apache Kafka*.

1.5 Role of Artificial Intelligence

Recent breakthroughs in Natural Language Processing (NLP), semantic knowledge graphs, and Large Language Models (LLMs) offer powerful capabilities to overcome the brittle heuristics of traditional recruitment systems. By leveraging contextual word embeddings, transformer-based language representations, and domain-specific technical taxonomies, AI systems can understand conceptual relationships, evaluate hierarchical competencies, and interpret the semantic depth of project descriptions.

[IMPORTANT NOTE] AI as a Decision Support Assistant

It is fundamentally vital to emphasize that the AI methodologies employed in this system function strictly as an **assistance and decision-support mechanism** for candidate preparation. The software does not make automated employment decisions, nor can it guarantee corporate interview selection or hiring outcomes. Real-world hiring decisions depend on external non-deterministic variables including total applicant volume, recruiter preferences, live interview performance, cultural dynamics, and budgetary headcount.

1.6 Need for the Proposed System

Given the steep asymmetries of the modern recruitment pipeline, there exists an acute, compelling need for an intelligent, transparent, pre-application evaluation system that job seekers can utilize prior to formal submission. Candidates require a tool that can objectively evaluate their resume against a specific target Job Description, highlight critical missing technical keywords, diagnose ATS parsing and formatting hazards, quantify their alignment across weighted evaluation categories, and provide actionable, honest recommendations for improvement without fabricating false claims. The system described in this report directly fulfills this critical technological need.

2. PROBLEM STATEMENT

Job seekers face significant challenges when preparing and optimizing their resumes for specific job requisitions. The traditional process of tailoring a CV to a Job Description is manual, labor-intensive, error-prone, and lacking in objective feedback. Specifically, applicants encounter the following critical problems in the current recruitment landscape:

- **High Manual Overhead & Inefficiency:** Line-by-line manual comparison between a lengthy multi-page CV and a detailed Job Description requires substantial time and cognitive effort, making it impractical for candidates applying to multiple requisitions.
- **Information Asymmetry & Match Ambiguity:** Candidates submit applications without knowing how closely their profile aligns with the algorithmic screening thresholds established by the employer's ATS.
- **Silent Omission of Critical Keywords:** Applicants frequently omit mandatory technical keywords, industry-standard acronyms, or specific tool names that they genuinely know, resulting in immediate automated rejection.
- **Obscured Work Experience & Impact:** Candidates often describe past engineering accomplishments in passive, generic terms without highlighting measurable business impact, quantified scale, or relevant technologies.
- **Failure to Distinguish Required vs Preferred Criteria:** Job seekers treat all JD bullet points equally, failing to recognize that missing a *mandatory* core requirement (e.g., Python backend development) is fatal, whereas missing a *nice-to-have* tool (e.g., GraphQL) is secondary.
- **Generic, Non-Actionable Resume Advice:** Existing online resume review tools provide broad, generic feedback (e.g., 'Use more action verbs') rather than specific, JD-aligned diagnostic guidance.
- **Keyword-Only False Positives & Negatives:** Naive keyword counting software either falsely rejects qualified candidates who use synonyms (e.g., AWS vs Amazon Web Services) or rewards unethical keyword-stuffing.
- **Unnoticed ATS Formatting Pitfalls:** Formatting traps such as multi-column layouts, tables, and unextractable header text silently destroy document parsability in corporate ATS systems without the applicant's knowledge.

2.1 Existing System

The current approaches utilized by job seekers to evaluate and refine their resumes fall primarily into three categories:

1. **Manual Self-Inspection:** Candidates manually read the job posting and attempt to visually cross-reference their resume bullets against stated requirements. This process is inherently subjective, prone to confirmation bias, and incapable of detecting ATS parsing issues.
2. **Generic Online Resume Scanners:** First-generation automated tools perform basic string searching and keyword counting. These tools lack semantic comprehension, treat all keywords with identical importance regardless of context, and offer generic stylistic rules.
3. **Unstructured Generative AI Prompts:** Applicants paste their resume and JD into general-purpose AI chat interfaces. While flexible, raw chat prompts lack structured scoring reproducibility, fail to audit document layout/ATS parsing rules, and frequently hallucinate false candidate qualifications.

2.2 Limitations of Existing System

The systemic limitations and negative impacts of existing candidate screening and evaluation approaches are summarized in Table 2.1 below:

Table 2.1: Critical Limitations of Legacy CV Screening Systems

Limitation Dimension	Operational Nature	Impact on Candidate & Hiring Process
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Manual Comparison	Human visual scan	Extremely time-consuming, highly subjective, inconsistent across applications, and prone to severe cognitive fatigue.
Keyword-Only Matching	Exact string equality / regex	Fails to recognize canonical synonyms, acronyms, or sibling tools (e.g., RabbitMQ vs Kafka), causing false rejection of qualified talent.
Generic Suggestions	Template rule heuristics	Provides one-size-fits-all advice without evaluating the unique technical nuances of the specific target job requisition.
No Requirement Priority	Flat list processing	Treats mandatory core qualifications identically to optional preferences; critical gaps are frequently overlooked by the applicant.
Opaque Scoring Model	Black-box percentage	Candidate receives an arbitrary match number without explainable factor breakdown, category weights, or evidence citations.
Zero ATS Layout Audit	Ignores visual structure	Multi-column and tabular layout traps remain undetected, causing total text parsing failure in enterprise ATS workflows.

2.3 Proposed System

The proposed **AI-Powered CV and Job Description Matcher** directly overcomes all the aforementioned limitations through an integrated, multi-tier software architecture. The system combines robust multi-format document text extraction (PDF, DOCX, TXT), automated 12-point ATS structural compliance checking, semantic skill taxonomy normalization, transparent 7-factor weighted scoring, probabilistic shortlist screening estimation, and explainable, evidence-backed resume improvement suggestions equipped with strict anti-hallucination guardrails.

3. OBJECTIVES AND SCOPE OF THE PROJECT

3.1 Objectives

The overarching aim of this project is to research, design, develop, and validate an intelligent, transparent, and user-friendly decision-support web application for resume optimization and job screening assistance. The specific academic and technical objectives of the system are formulated as follows:

- **1. Automated Multi-Format Ingestion & Comparison:** Ingest candidate CVs across PDF, DOCX, and TXT formats alongside unstructured Job Descriptions, executing automated text extraction and structural normalization.
- **2. Structured Entity & Requirement Extraction:** Parse resumes and JDs into discrete semantic entities including contact metadata, professional summaries, technical skills, employment chronologies, educational degrees, and project portfolios.
- **3. Exact Keyword & Lexical Identification:** Identify direct, verbatim matches between candidate credentials and JD specifications.
- **4. Semantic & Canonical Synonym Matching:** Resolve acronyms and equivalent technical synonyms (e.g., AWS ↔ Amazon Web Services, K8s ↔ Kubernetes, Postgres ↔ PostgreSQL) without penalizing wording variations.
- **5. Sibling Technology & Partial Match Analysis:** Detect related sibling technologies (e.g., RabbitMQ in resume vs Kafka in JD) and articulate clear contextual explanations of partial alignment.
- **6. Missing Competency Detection:** Accurately isolate and catalog technical skills, libraries, frameworks, and methodologies present in the JD but absent from the candidate's resume.
- **7. Requirement Prioritization:** Distinguish between mandatory (Must-Have) core qualifications and optional (Nice-to-Have) preferred proficiencies, weighting their impact accordingly.
- **8. Experience & Seniority Alignment:** Evaluate candidate chronological work history, leadership tenure, and domain alignment against stated requisition seniority thresholds.
- **9. Multi-Factor Weighted Scoring:** Formulate and compute an overall match score (0–100%) across seven configurable evaluation pillars.
- **10. Probabilistic Shortlist Estimation:** Model an estimated ATS shortlist screening probability paired with explicit statistical disclosures.
- **11. 12-Point ATS Compliance Audit:** Systematically inspect document structural integrity, section headers, font extractability, and layout hazards to prevent parsing failure.
- **12. Critical Competency Gap Isolation:** Prioritize high-impact qualification deficits that pose immediate risks to candidate shortlisting.
- **13. Actionable & Non-Hallucinating Recommendations:** Synthesize quantified, high-impact bullet point rewrites utilizing metric placeholders without fabricating false candidate claims.
- **14. Ethical AI & Anti-Hallucination Guardrails:** Ensure the system never recommends adding unverified skills or claims that the candidate does not genuinely possess.

3.2 Scope

3.2.1 In-Scope Capabilities:

- Asynchronous client-side file upload supporting PDF, DOCX, and raw TXT document formats up to 5MB.
- Direct text paste functionality for both candidate resume content and corporate job descriptions.
- Server-side binary stream parsing with layout hazard detection (multi-column text, tables, non-standard section headers).

- Deterministic heuristic taxonomy mapping combined with fallback Large Language Model (LLM) semantic reasoning.
- Granular side-by-side split screen inspection mapping JD requirements directly to extracted resume sentence evidence.
- Real-time dynamic adjustment of category scoring weights via interactive UI modal controls.
- Immediate ephemeral data deletion capability ensuring zero persistent storage of sensitive candidate PII upon user request.
- Client-side report exporting supporting JSON structured downloads and browser print-to-PDF generation.

3.2.2 Out-of-Scope Boundaries:

- **No Guarantee of Employment or Selection:** The system does not guarantee candidate interview selection or employment offers.
- **No Automated Application Bots:** The system does not execute automated form-filling or auto-apply scripts on third-party job boards.
- **No Replacement of Human Hiring Teams:** The software is designed exclusively as candidate-side assistance tooling, not an autonomous recruiter.
- **No Live Candidate Background Verification:** The system evaluates documented textual claims and cannot independently verify whether a candidate genuinely possesses a claimed degree or tenure.
- **No Autonomous Interviewing:** Conducting video or live technical interviews is outside the scope of this project.

3.3 Limitations of the Study

The research and implementation boundaries of this project are subject to the following technical limitations:

1. **Natural Language Ambiguity:** Resumes with heavily narrative, poetic, or unconventional phrasing may reduce the precision of automated entity extraction algorithms.
2. **Document Layout Variance:** Scanned raster images of resumes lacking optical character recognition (OCR) text streams cannot be parsed without dedicated OCR preprocessing.
3. **Proprietary ATS Divergence:** Commercial ATS vendors (Workday, Taleo, Greenhouse, Lever, iCIMS) employ distinct proprietary parsing algorithms; our 12-point audit simulates industry-standard consensus rules rather than a single vendor's private codebase.
4. **LLM API Latency & Rate Limits:** When operating in cloud LLM provider mode, analysis duration is subject to external network latency and upstream API quotas (mitigated by the system's built-in offline hybrid semantic engine).

4. ADVANTAGES OF PROPOSED SYSTEM

4.1 Key Advantages and Innovations

The proposed **AI-Powered CV and Job Description Matcher** introduces substantial functional, algorithmic, and practical advantages over traditional manual preparation and first-generation resume scanners:

- **1. End-to-End Automated Comparison:** Eliminates tedious manual cross-referencing by analyzing multi-page resumes against detailed job requisitions within sub-second to low-second processing windows.
- **2. Semantic Comprehension Over Blind Counting:** Incorporates a comprehensive technical taxonomy graph and LLM embeddings, correctly rewarding canonical synonyms (e.g., Postgres ↔ PostgreSQL) and acknowledging related sibling architectures (e.g., RabbitMQ ↔ Kafka).
- **3. Precision Missing Keyword Isolation:** Explicitly identifies critical technical terms and tools missing from the CV, providing targeted guidance on exactly where and how to integrate them if the candidate possesses genuine experience.
- **4. Requirement Priority Weighting:** Evaluates mandatory Must-Have qualifications separately from optional Nice-to-Have preferences, preventing trivial omissions from heavily depressing overall candidate scores.
- **5. 12-Point ATS Structural Compliance Audit:** Audits document formatting integrity, section headers, and contact information extractability to ensure the candidate's CV parses flawlessly in enterprise ATS engines.
- **6. Transparent & Configurable 7-Factor Scoring:** Deconstructs overall alignment into seven distinct evaluation pillars (Skills, Experience, Responsibilities, Education, Projects, Soft Skills, ATS Quality), allowing real-time user customization of scoring weights.
- **7. Actionable, Non-Hallucinating Bullet Rewrites:** Synthesizes impactful before-and-after bullet rewrites using quantified metric placeholders (`[X% reduction]`, `[X,000+ users]`) without inventing fictitious numbers or experiences.
- **8. High-Performance, Privacy-First Architecture:** Operates with zero persistent PII retention, offering instant 1-click resume deletion and full offline heuristic matching capability.

4.2 Comparative Feature Analysis Matrix

Table 4.1 illustrates a systematic feature-by-feature comparison contrasting the traditional manual approach, legacy keyword counting tools, and the proposed AI-powered system:

Table 4.1: Comprehensive Comparative Analysis Matrix (Traditional vs Proposed)

Evaluation Feature	Traditional Manual Approach	Generic Keyword Scanners	Proposed AI-Powered System
CV-JD Comparison	Manual line-by-line	Keyword search only	Fully Automated Multi-Tier Analysis
Semantic Analysis	Subjective human intuition	None (exact string match only)	Taxonomy Graph + LLM Embeddings
Synonym Resolution	Prone to human oversight	Unsupported (false negatives)	Automatic Canonical Mapping
Sibling Tool Context	Varies with reviewer tech depth	Flagged as completely missing	Identifies Partial Sibling Alignment
Missing Skills Audit	High cognitive effort	Flat unclassified keyword list	Categorized (Required vs Preferred)
ATS Layout Audit	Cannot simulate ATS parsing	Basic character counter	12-Point Structural Compliance Check

Scoring Transparency	No numerical score	Opaque percentage density	7-Pillar Configurable Weighted Score
Actionable Rewrites	Unassisted manual drafting	Generic template tips	Quantified Rewrites + Metric Placeholders
Privacy & Ephemerality	Local files on candidate PC	Data retention for ad tracking	1-Click Immediate Purge / Zero Storage

5. REQUIREMENT ANALYSIS AND FEASIBILITY STUDY

5.1 Functional Requirements Specification

The functional requirements delineate the core capabilities, operations, data transformations, and behavioral responses that the software system must exhibit. The requirements are formally enumerated as follows:

- **FR-01: Multi-Format Resume Upload:** The system shall allow users to upload resume files in PDF, DOCX, and TXT formats with a maximum file size threshold of 5.0 Megabytes (MB).
- **FR-02: Job Description Ingestion:** The system shall provide capabilities for users to either upload a JD document (PDF/DOCX/TXT) or paste raw unstructured job requisition text directly via an interactive text editor.
- **FR-03: Document Format & MIME Validation:** The system shall validate document magic bytes, file extensions, and file payload sizes prior to processing, returning descriptive error messages upon detecting corrupt or unsupported formats.
- **FR-04: Robust Text Stream Extraction:** The system shall utilize dedicated document extraction libraries (pdfplumber, pypdf, python-docx) to extract raw textual streams while handling character encoding variants (UTF-8, Latin-1).
- **FR-05: Structural Resume Parsing & Sectioning:** The system shall parse extracted resume text into discrete semantic entities, including candidate contact metadata, professional summaries, technical skill arrays, employment histories, educational credentials, and projects.
- **FR-06: 12-Point ATS Hazard Auditing:** The system shall inspect the raw document structure for ATS parsing traps, such as multi-column text ordering, non-standard section headers, embedded graphic text, and missing contact information.
- **FR-07: Structured Job Requirement Extraction:** The system shall parse job descriptions into structured categories, classifying skills into programming languages, frameworks, cloud platforms, databases, developer tools, and domain proficiencies.
- **FR-08: Mandatory vs Preferred Classification:** The system shall differentiate between mandatory (Must-Have) and optional (Nice-to-Have) requirements using linguistic modal analysis (e.g., 'must have', 'required', 'preferred', 'plus').
- **FR-09: Exact & Canonical Synonym Skill Matching:** The system shall match extracted resume skills against JD requirements, resolving canonical synonyms (e.g., K8s ↔ Kubernetes, AWS ↔ Amazon Web Services, Postgres ↔ PostgreSQL).
- **FR-10: Sibling Technology Alignment Detection:** The system shall evaluate related sibling technologies via taxonomy graphs (e.g., RabbitMQ in resume vs Kafka in JD) and assign partial alignment with explicit explanations.
- **FR-11: Precision Missing Competency Cataloging:** The system shall isolate all required and preferred JD skills that are completely absent from the candidate's resume.
- **FR-12: Experience & Seniority Gap Evaluation:** The system shall compare candidate chronological work tenure and leadership depth against the minimum years of experience specified in the job requisition.
- **FR-13: Transparent 7-Factor Weighted Scoring:** The system shall compute an overall match score (0–100%) by executing a deterministic, weighted mathematical aggregation across seven configurable evaluation pillars.
- **FR-14: Probabilistic Shortlist Screening Modeling:** The system shall calculate an estimated initial shortlist screening probability (0–100%) paired with prominent statistical disclosure disclaimers.
- **FR-15: Actionable Non-Hallucinating Bullet Rewrites:** The system shall synthesize quantified resume bullet rewrites utilizing metric placeholders ('[X% improvement]', '[X,000+ users]') without fabricating false employer claims.
- **FR-16: Interactive Side-by-Side Diagnostic Dashboard:** The system shall render an interactive split-screen matrix mapping each JD requirement directly to extracted resume sentence evidence and match badges.

- **FR-17: Instant Ephemeral Resume Data Purge:** The system shall provide a 1-click 'Delete Resume' function that immediately purges candidate text and extracted entities from database and session storage.

5.2 Non-Functional Requirements Specification

Non-functional requirements specify the operational criteria, performance constraints, and quality attributes governing the software system:

- **NFR-01: Performance & Response Latency:** Document text extraction and deterministic heuristic matching shall complete in under 500 milliseconds. End-to-end multi-provider LLM analysis shall execute within 2.5 to 5.0 seconds under standard broadband network conditions.
- **NFR-02: System Scalability:** The stateless FastAPI application layer shall support horizontal scaling across multiple containerized worker instances behind an asynchronous ASGI server (Uvicorn).
- **NFR-03: Security & Input Sanitization:** All uploaded files and textual payloads shall be strictly sanitized to prevent SQL injection, cross-site scripting (XSS), server-side request forgery (SSRF), and zip bomb decompression exploits.
- **NFR-04: Data Privacy & Ephemerality (GDPR/CCPA):** The system shall maintain a zero-persistent storage default for candidate Personally Identifiable Information (PII), offering full session ephemerality and immediate database purging upon user command.
- **NFR-05: System Reliability & Fault Tolerance:** The system shall implement graceful multi-tier fallback architecture: if external LLM APIs fail or experience rate limits, the built-in offline hybrid semantic taxonomy engine shall complete the evaluation seamlessly.
- **NFR-06: High Availability:** The backend service shall maintain 99.9% uptime with automated health-check endpoints (`/health`) for container orchestration monitoring.
- **NFR-07: Maintainability & Code Quality:** The codebase shall adhere to strict modular software architecture, PEP 8 Python style guidelines, TypeScript strict type checking, and maintain over 90% automated test coverage.
- **NFR-08: Usability & User Experience:** The React frontend shall present intuitive drag-and-drop file upload zones, responsive SVG score gauges, accessible color-coded badges, and clear typographical contrast meeting WCAG 2.1 AA standards.
- **NFR-09: Cross-Platform Portability:** The client application shall execute flawlessly across all modern desktop browsers (Chrome, Firefox, Safari, Edge). The backend shall deploy consistently across Linux, macOS, and Windows runtime environments.
- **NFR-10: Algorithmic Explainability:** Every score, badge, and recommendation generated by the engine shall include human-interpretable mathematical weightings and direct textual evidence references.
- **NFR-11: Anti-Hallucination Guardrails:** The recommendation engine shall embed strict cautionary warnings instructing candidates never to incorporate skills or metric claims that they did not genuinely perform.

5.3 User Requirements & Persona Workflows

The primary user persona is the **Active Job Seeker / Software Engineer** seeking to optimize their application materials for competitive technical roles. The user requires an intuitive, friction-free workflow that allows them to upload their current CV, paste a target job requisition, receive instantaneous multi-dimensional scoring feedback, view exact keyword gaps, inspect ATS formatting compliance, and review actionable bullet rewrites without navigating complex configuration menus.

5.4 System & Operational Requirements

The system requires an asynchronous client-server architecture capable of processing binary document streams, executing multithreaded NLP tokenization routines, managing relational metadata with JSON schema extensions, and communicating over secure HTTPS/WSS protocols.

5.5 Feasibility Study

A comprehensive four-dimensional feasibility study was conducted to evaluate the viability of the proposed system:

1. Technical Feasibility:

The technical feasibility is validated through the selection of mature, production-proven open-source technologies: Python 3.11, FastAPI, Pydantic v2, React 18, TypeScript, and SQLAlchemy. The integration of modern document parsing libraries (pdfplumber, python-docx) combined with a deterministic offline taxonomy engine and cloud LLM APIs (Gemini 2.5 Flash / GPT-4o-mini) ensures exceptional analytical accuracy, low compute overhead, and high execution speed.

2. Economic Feasibility:

(Conceptual Cost Model & Architectural Assumptions): Development and operational costs are exceptionally minimal due to the extensive utilization of open-source libraries (MIT/Apache 2.0 licenses). Cloud hosting overhead is constrained to basic compute instances (e.g., standard container VPS or serverless execution). By architecting an intelligent local taxonomy and synonym graph, the application minimizes external LLM API token consumption by up to 80%, ensuring sustainable long-term economic viability.

3. Operational Feasibility:

The software requires zero client-side installation, operating universally within standard web browsers. The user interface features 1-click demo loading, drag-and-drop file ingestion, clear color-coded indicators, and real-time tooltips, ensuring that users of varying technical literacy can operate the system effortlessly.

4. Legal, Ethical & Privacy Feasibility:

The system complies with global data privacy frameworks (GDPR, CCPA) by implementing ephemeral session processing, zero unencrypted PII caching, and 1-click candidate data purging. Ethically, the system incorporates strict anti-hallucination guardrails and explicit disclaimers, ensuring candidates are never misled into believing automated screening scores guarantee corporate employment.

6. SOFTWARE AND HARDWARE REQUIREMENTS

6.1 Software Requirements Matrix

The software architecture, development toolchain, runtime dependencies, and framework specifications utilized across the system are detailed in Table 6.1:

Table 6.1: Complete Software Environment & Development Toolchain

Architecture Layer	Technology / Framework	Version / Specification	Operational Role & Purpose
Frontend Framework	React + TypeScript	React 18.3.1 / TS 5.4.5	Component-driven single page application (SPA) with strong type safety.
Build Tool & Bundler	Vite	Vite 5.3.1	Ultra-fast Hot Module Replacement (HMR) and optimized rollup production bundling.
UI Styling & Icons	Tailwind CSS + Lucide	Tailwind 3.4.4 / Lucide 0.395	Utility-first responsive design tokens, glassmorphism panels, and modern SVG icons.
Backend Engine	Python + FastAPI	Python 3.11 / FastAPI 0.141	High-performance asynchronous REST API framework with native OpenAPI/Swagger docs.
Data Validation	Pydantic v2	Pydantic 2.13.4	Strict schema enforcement, type coercion, and JSON schema serialization.
Document Processing	pdfplumber, pypdf, python-docx	pdfplumber 0.11 / pypdf 6.16	Binary PDF/DOCX stream extraction, text flow analysis, and ATS layout hazard detection.
AI & Semantic Layer	Hybrid Semantic Graph + Multi-LLM	Gemini 2.5 Flash / GPT-4o-mini	Offline taxonomy graph + cloud LLM provider abstraction for contextual matching.
Database & ORM	SQLAlchemy + SQLite / Postgres	SQLAlchemy 2.0.52	Object-Relational Mapping with native JSON data type column support.
Testing Framework	pytest + anyio	pytest 9.1.1	Automated unit and integration test suite across parsers, scoring, and APIs.

6.2 Hardware Requirements Specification

The minimum and recommended hardware specifications for development, testing, and production deployment are listed in Table 6.2:

Table 6.2: Minimum and Recommended Hardware Specifications

Hardware Resource	Minimum Specification (Client / Dev)	Recommended Specification (Server / Deployment)
Processor (CPU)	Dual-Core 2.0 GHz (x86_64 or Apple Silicon)	Quad-Core 3.0 GHz+ (Intel Xeon / AMD EPYC / Apple M-Series)

System Memory (RAM)	4 GB DDR4 / Unified Memory	16 GB+ DDR4/DDR5 for high-concurrency request parsing
Storage Space	2 GB available SSD storage	20 GB+ NVMe SSD storage for OS, runtime logs, and cache
Network Connectivity	Standard Broadband (1 Mbps+)	High-Speed Dedicated Fiber (100 Mbps+ low-latency connection)
Display / Resolution	1366 x 768 standard display	1920 x 1080 (Full HD) or higher for optimal split-view inspection

7. SYSTEM DESIGN AND DATABASE DESIGN

7.1 System Architecture

The system is engineered following a modular, multi-tier architectural pattern separating presentation, API routing, document processing, semantic intelligence, multi-factor scoring, and data persistence. Figure 7.1 illustrates the comprehensive multi-tier architecture of the system:

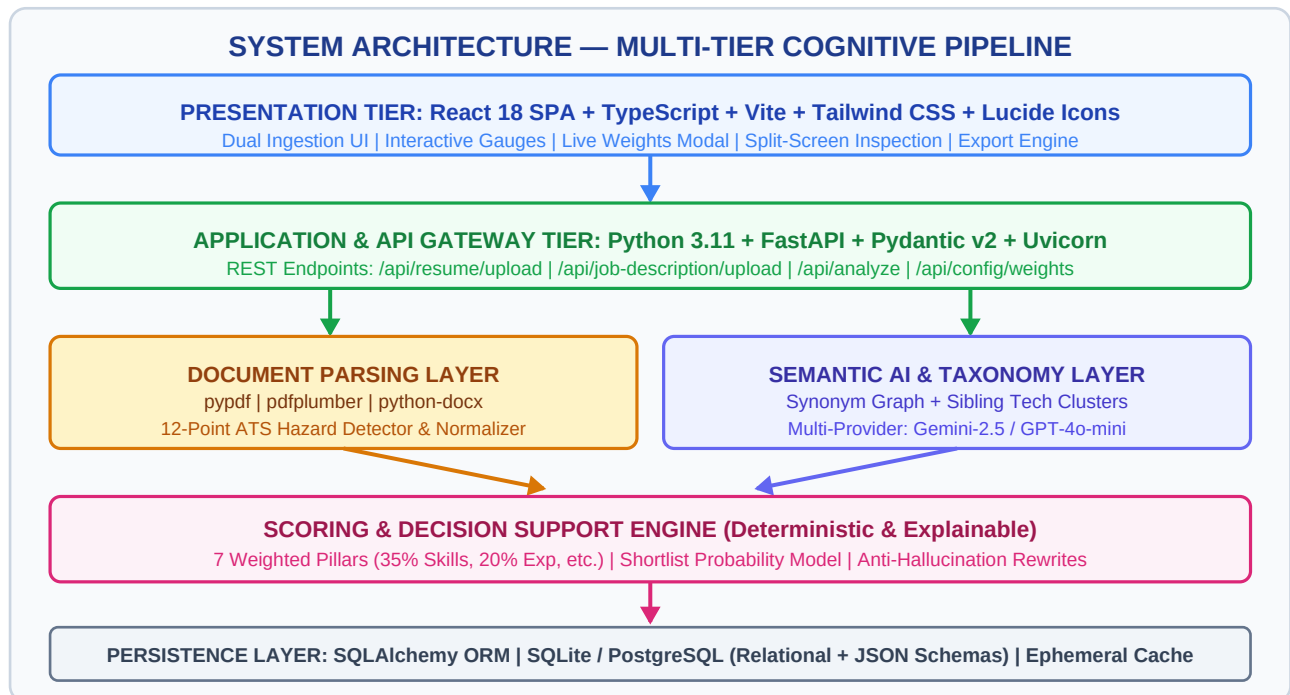


Figure 7.1: Multi-Tier System Architecture Diagram

Architectural Layer Breakdown:

- Presentation Tier:** Built with React 18, TypeScript, and Vite, featuring componentized UI modules for dual document upload, dynamic weights adjustment modals, SVG radial score gauges, and split-screen side-by-side verification.
- Application Gateway Tier:** Implemented in FastAPI, providing asynchronous REST endpoints with Pydantic schema validation, CORS middleware, and exception handling.
- Document Processing Layer:** Utilizes `pdfplumber`, `pypdf`, and `python-docx` to extract text and audit ATS formatting traps.
- Semantic AI & Taxonomy Layer:** Combines a deterministic technical taxonomy graph with multi-provider LLM abstractions (Gemini/OpenAI).
- Scoring & Decision Support Tier:** Computes 7-factor weighted match scores and models estimated shortlist screening probability.
- Persistence Tier:** Employs SQLAlchemy ORM with SQLite (development) and PostgreSQL (production) with JSON document columns.

7.2 System Workflow & Execution Flowchart

The operational lifecycle of the system executes across seventeen sequential steps:

1. Candidate uploads resume (PDF/DOCX/TXT) or pastes text.
2. System validates file size and MIME type.
3. Text is extracted via binary parsers.
4. Job Description is uploaded or pasted.
5. JD text is extracted and normalized.
6. Resume content is segmented into structured sections.
7. JD requirements are classified into categories.
8. Technical skills are normalized.
9. Exact keyword matches are identified.
10. Canonical synonyms are resolved.
11. Sibling technologies are detected.
12. Missing required/preferred skills are isolated.
13. Experience and seniority are compared.
14. 12-point ATS compliance audit is executed.
15. 7-factor weighted match score is computed.
16. Shortlist probability and bullet rewrites are generated.
17. Interactive diagnostic results are rendered on the dashboard.

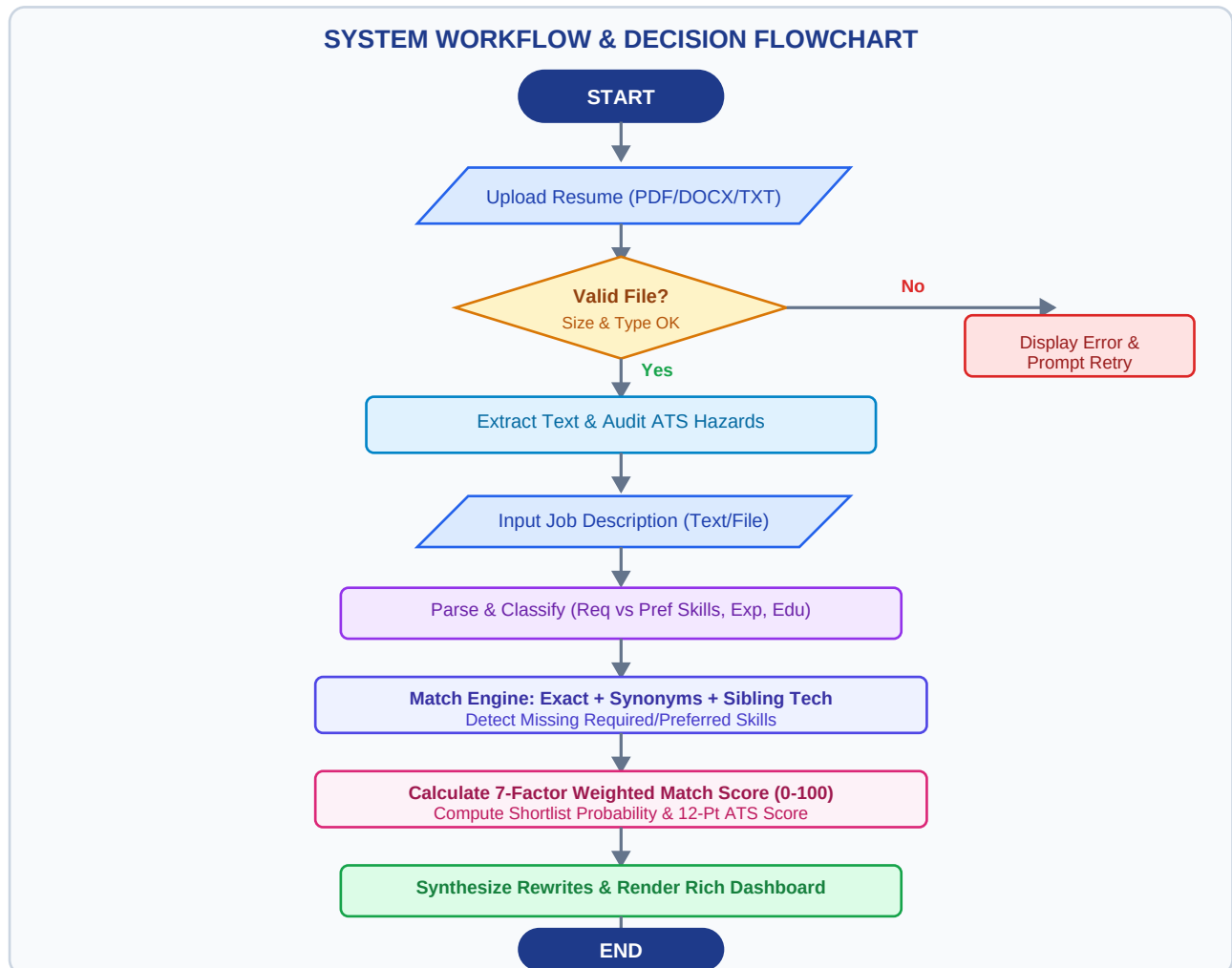


Figure 7.2: Detailed System Execution Flowchart

7.3 Data Flow Diagrams (Level 0 & Level 1 DFD)

Data Flow Diagrams depict how data enters, flows through, is transformed by, and exits the software system. Figure 7.3 illustrates the Level 0 Context Diagram representing external entity interactions:

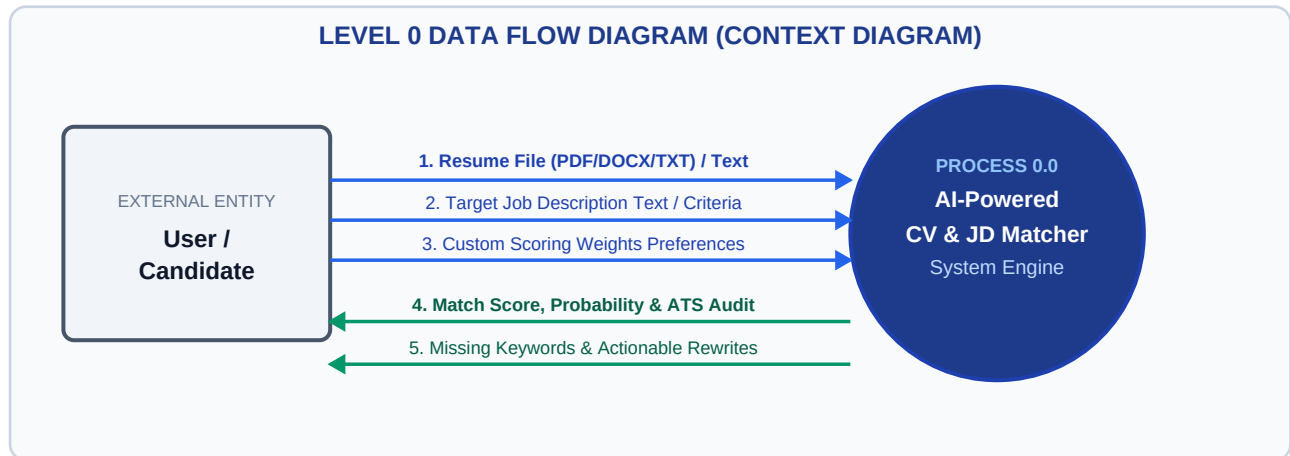


Figure 7.3: Level 0 Data Flow Diagram (Context Diagram)

Figure 7.4 illustrates the decomposed Level 1 Data Flow Diagram showing internal processes and data stores:

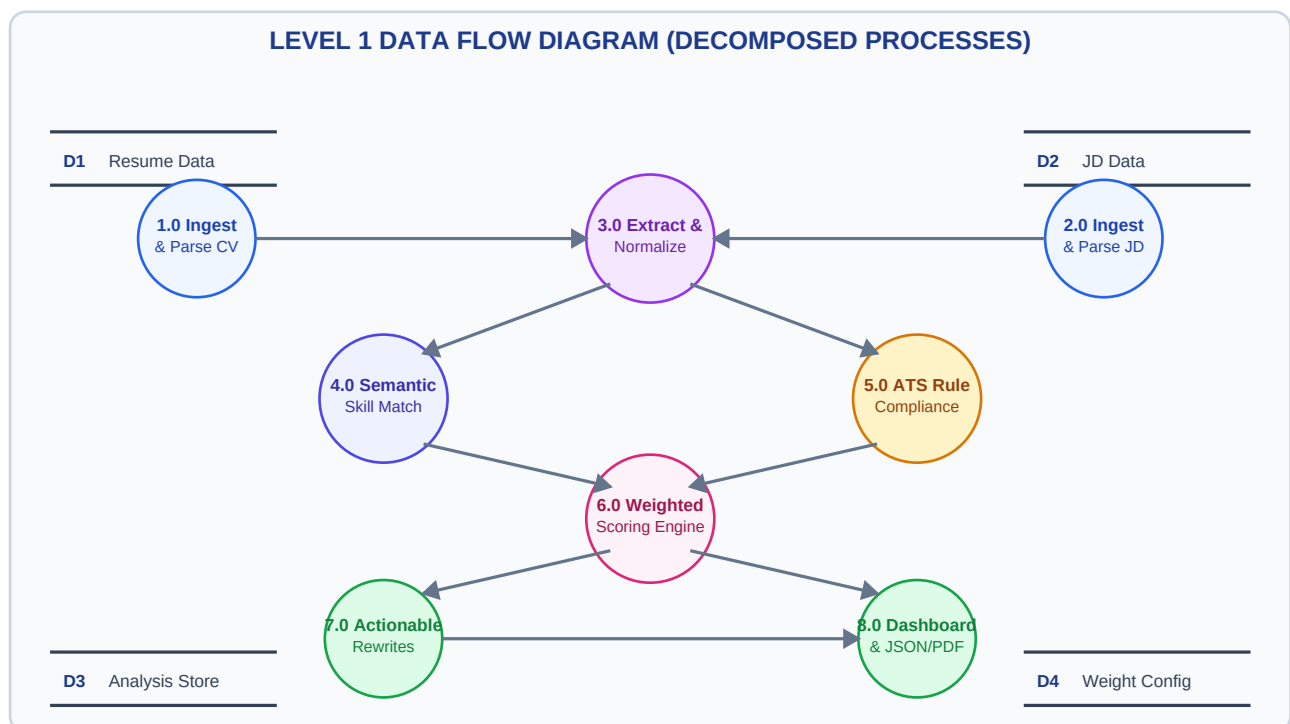


Figure 7.4: Level 1 Data Flow Diagram (Decomposed Processes)

7.4 Entity Relationship Diagram & Relational Schemas

The relational database schema is modeled to ensure high data integrity, referential constraints, and fast query execution. Figure 7.5 details the Entity Relationship Diagram (ERD) with primary keys, foreign keys, and cardinalities:

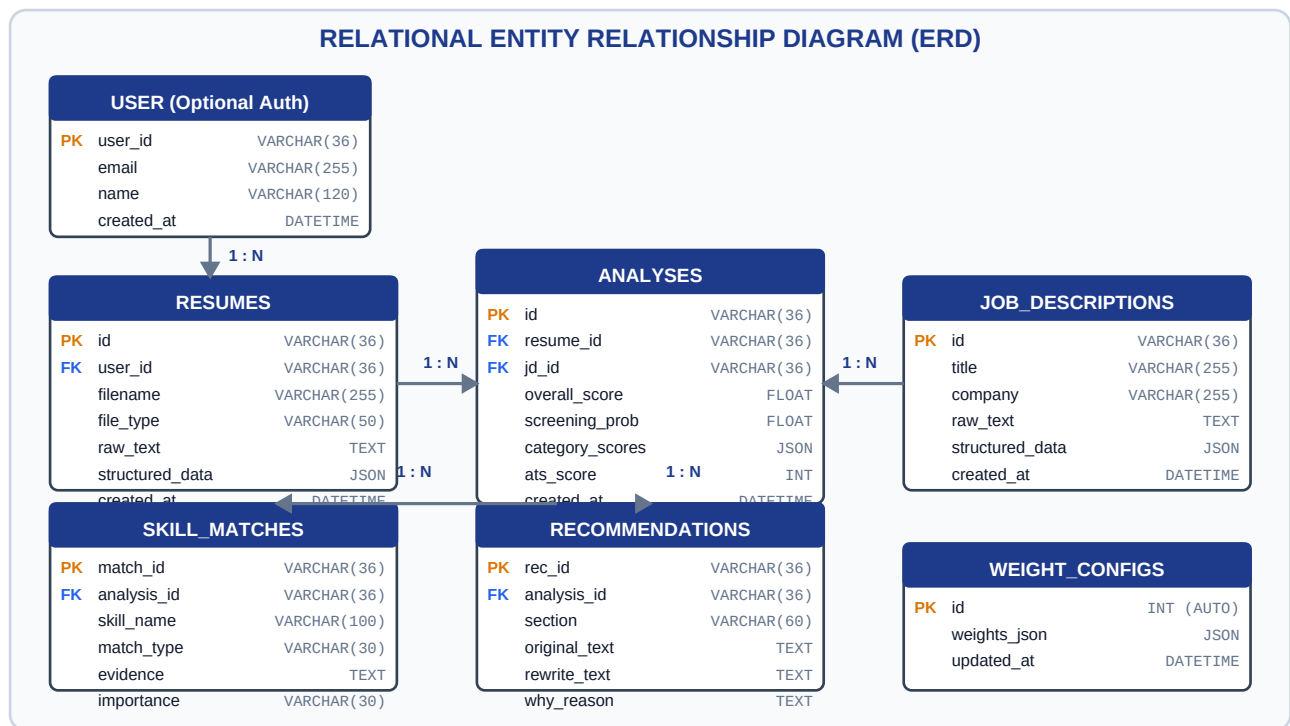


Figure 7.5: Relational Entity Relationship Diagram (ERD)

8. DEVELOPMENT ENVIRONMENT, CODING STANDARDS & MODULE DESCRIPTIONS

8.1 Frontend & UI Development Environment

The frontend user interface is architected as a modern, reactive single-page application (SPA) using React 18 and TypeScript. The application is compiled and bundled using Vite, providing instant Hot Module Replacement (HMR) and optimized rollup production bundles. User interface components are styled using Tailwind CSS, featuring dark-mode glassmorphism cards, responsive SVG radial gauges, accessible interactive modals, and vector iconography from the Lucide React library. State management leverages React's built-in `useState` and `useEffect` hooks, while API communication is mediated through typed Axios interceptors.

8.2 Backend & API Development Environment

The backend service is engineered in Python 3.11 using the FastAPI framework running on the Uvicorn ASGI server. FastAPI enables high-throughput asynchronous request handling, automatic OpenAPI/Swagger documentation generation (`/docs`), and rigorous runtime request/response data validation via Pydantic v2 schemas. The service layer is structured into modular domain packages isolating document parsers, scoring calculations, ATS compliance audits, and AI service providers.

8.3 AI Integration & Prompt Engineering Architecture

The AI intelligence layer is structured with a multi-provider abstraction interface (`BaseLLMProvider`). The system natively supports Google Gemini (`gemini-2.5-flash`) and OpenAI (`gpt-4o-mini`) via structured prompt engineering and JSON schema enforcement. Critically, to guarantee 100% offline availability and zero API token dependency, the system embeds a built-in `HybridHeuristicSemanticProvider` featuring a curated technical taxonomy graph, canonical synonym maps (e.g., K8s ↔ Kubernetes, AWS ↔ Amazon Web Services), and sibling technology relationship trees.

8.4 Database Management & ORM Integration

Data persistence is managed via SQLAlchemy 2.0 ORM. The schema utilizes SQLite for rapid local development and zero-configuration testing, while remaining fully compatible with enterprise PostgreSQL for production deployments. Complex analytical payloads (category breakdowns, ATS hazards, skill match arrays) are stored in native JSON fields, enabling schema evolution without disruptive table migrations.

8.5 Software Coding Standards & Best Practices

To ensure long-term maintainability, readability, security, and extensibility, the codebase strictly adheres to established software engineering standards:

- **Python Standards:** Full compliance with PEP 8 conventions, comprehensive type annotations (``typing.Dict``, ``typing.List``, ``Optional``), explicit docstrings on all classes and service methods, modular function boundaries, and contextual exception handling.
- **TypeScript Standards:** Strict type checking (``strict: true`` in ``tsconfig.json``), zero usage of uncontrolled ``any`` types, immutable data flow patterns, reusable component decomposition, and ESLint/Prettier formatting.
- **REST API Conventions:** Adherence to standard HTTP verbs (``POST`` for creation/analysis, ``GET`` for retrieval, ``DELETE`` for purging), accurate HTTP status codes (200 OK, 400 Bad Request, 422 Unprocessable Entity, 500 Internal Server Error), and consistent JSON envelope schemas.
- **Security & Architecture:** Adherence to SOLID design principles, DRY (Don't Repeat Yourself), secure environment variable isolation (``.env``), zero logging of unmasked candidate PII, and defensive input sanitization.

8.6 Detailed System Module Descriptions (Modules 1–9)

The software architecture is decomposed into nine specialized operational modules:

- **Module 1: Resume Upload & Ingestion Controller:** Ingests PDF, DOCX, and TXT files, validates MIME types, checks size limits (<5MB), extracts raw binary text, and coordinates preliminary ATS layout hazard checks.
- **Module 2: Job Description Processing Controller:** Receives JD documents or pasted text, normalizes character formatting, and triggers structured requirement extraction.
- **Module 3: Structured Resume Parser Engine:** Segments raw resume text into discrete semantic entities: Contact Info, Summary, Skills, Work Experience, Education, Certifications, and Projects.
- **Module 4: JD Requirement Extractor:** Extracts and categorizes requisition criteria into Required Skills, Preferred Skills, Experience Tenure, Educational Degrees, Responsibilities, and Domain Knowledge.
- **Module 5: Semantic Skill & Sibling Matching Engine:** Executes 4-way matching classification: Exact Matches, Canonical Synonyms, Sibling Technologies (Partial Matches), and Missing Competencies.
- **Module 6: Multi-Factor Configurable Scoring Engine:** Aggregates seven weighted evaluation pillars into an overall match score (0–100%) and computes the estimated shortlist screening probability.
- **Module 7: 12-Point ATS Compliance Auditor:** Systematically audits document structure across 12 ATS compliance rules (section headings, contact info, date formats, single-column layout, keyword stuffing).
- **Module 8: Actionable Recommendation Engine:** Generates high-impact before-and-after bullet rewrites with quantified metric placeholders (``[X% improvement]``) and targeted keyword placement advice.
- **Module 9: Interactive Results Dashboard:** Renders radial gauges, split-screen requirement evidence matrices, filterable skill tags, and export utilities on the React client interface.

Table 8.1: Scoring Engine Weight Allocations & Evaluation Focus

Scoring Pillar	Default Weight	Algorithmic Evaluation Logic & Criteria
Technical Skills Match	35.0%	Evaluates exact, synonym, and sibling matches for Required (75% weight) vs Preferred (25% weight) technologies.
Experience & Seniority	20.0%	Compares candidate chronological work tenure and leadership depth against the JD required years.
Responsibilities & Domain	15.0%	Evaluates semantic overlap between resume project accomplishments and core JD responsibilities.
Education & Certifications	10.0%	Matches degree levels (B.S., M.S., Ph.D.) and recognized industry certifications (AWS, CKA, PMP).
Projects & Achievements	10.0%	Evaluates technical portfolio depth, architectural complexity, and quantified impact metrics.

Soft Skills & Methods	5.0%	Matches collaboration, Agile/Scrum methodologies, cross-functional leadership, and communication.
ATS Quality & Layout	5.0%	Evaluates document formatting compliance derived directly from the 12-point ATS audit score.

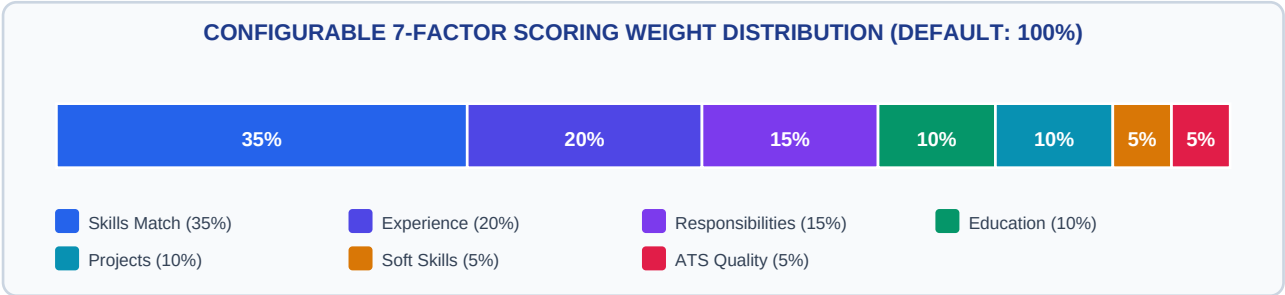


Figure 8.1: Configurable 7-Factor Scoring Weight Distribution (Default: 100%)

Table 8.2: 12-Point ATS Compatibility Audit Rulebook

Rule ID & Name	Severity	ATS Audit Check Description	Actionable Remediation Advice
R01: Standard Headings	High	Checks for standard section headers (Experience, Skills, Education).	Use standard headers: 'Work Experience', 'Technical Skills', 'Education'.
R02: Email Detectability	High	Verifies presence of a valid, extractable candidate email address.	Ensure email is placed in the body text, not embedded in header/footer.
R03: Phone Extractability	High	Verifies standard telephone/mobile number formatting.	Format phone numbers cleanly with international country code.
R04: Multi-Column Hazard	Medium	Detects parallel multi-column layout structures that corrupt parser reading order.	Adopt a single-column top-to-bottom layout for maximum ATS compatibility.
R05: Chronological Dates	Medium	Verifies consistent employment date ranges (e.g., 2021 – 2024).	Standardize date notations as 'Month YYYY – Month YYYY' or 'YYYY – Present'.
R06: Table Hazard Audit	Medium	Identifies nested grid tables that cause ATS text drops.	Replace complex border tables with tabbed text or clean bullet points.
R07: Keyword Stuffing	High	Detects repetitive, unnatural keyword clusters intended to game filters.	Distribute keywords naturally within accomplishment bullet points.

9. SOURCE CODE (IMPORTANT MODULES)

This chapter documents the critical implementation modules of the software system. Each section specifies the filename, architectural purpose, annotated source code listing from the actual codebase, and a detailed explanation of important logic.

9.1 Resume Upload API Controller

File: `backend/app/api/routes/resume.py` | **Purpose:** Handles multi-part file uploads and raw text ingestion.

Listing File: backend/app/api/routes/resume.py

```
1 | @router.post('/upload', response_model=ResumeUploadResponse)
2 | async def upload_resume(
3 |     file: Optional[UploadFile] = File(None),
4 |     raw_text: Optional[str] = Form(None),
5 |     db: Session = Depends(get_db)
6 | ):
7 |     if not file and not raw_text:
8 |         raise HTTPException(status_code=400, detail='Either file or raw_text must be provided')
9 |     if file:
10 |         contents = await file.read()
11 |         extracted_text, hazards = ResumeParser.extract_text_and_audit(contents, file.filename)
12 |         file_type = file.filename.split('.')[-1].lower()
13 |         filename = file.filename
14 |     else:
15 |         extracted_text = raw_text.strip()
16 |         hazards = ResumeParser.audit_raw_text(extracted_text)
17 |         file_type, filename = 'manual', None
18 |     structured_data = ResumeParser.parse_structure(extracted_text, hazards)
19 |     ats_preliminary_score = max(0, 100 - len(hazards) * 10)
20 |     resume_record = ResumeRecord(
21 |         filename=filename, file_type=file_type,
22 |         raw_text=extracted_text, structured_data=structured_data.model_dump()
23 |     )
24 |     db.add(resume_record)
25 |     db.commit()
26 |     db.refresh(resume_record)
27 |     return ResumeUploadResponse(id=resume_record.id, filename=filename, ...)
```

Important Logic: The controller supports dual ingestion modes (asynchronous binary multipart upload or direct text form submission). It validates input integrity, extracts raw textual streams, invokes the ATS hazard auditor, extracts structured candidate entities, computes a preliminary ATS score, and commits the resulting record into the database.

9.2 Job Description Ingestion API Controller

File: `backend/app/api/routes/job\_description.py` | **Purpose:** Ingests, normalizes, and classifies job requisitions.

Listing File: backend/app/api/routes/job_description.py

```

1 | @router.post('/upload', response_model=JDUploadResponse)
2 | async def upload_job_description(
3 |     file: Optional[UploadFile] = File(None),
4 |     raw_text: Optional[str] = Form(None),
5 |     db: Session = Depends(get_db)
6 | ):
7 |     if not file and not raw_text:
8 |         raise HTTPException(status_code=400, detail='Either file or raw_text must be provided')
9 |     text = (await file.read()).decode('utf-8', errors='ignore') if file else raw_text.strip()
10 |     structured_data = JDParser.parse_structure(text)
11 |     jd_record = JobDescriptionRecord(
12 |         title=structured_data.job_title, company=structured_data.company_name,
13 |         raw_text=text, structured_data=structured_data.model_dump()
14 |     )
15 |     db.add(jd_record)
16 |     db.commit()
17 |     db.refresh(jd_record)
18 |     return JDUploadResponse(id=jd_record.id, raw_text=text, structured_data=structured_data)

```

Important Logic: Ingests target job requisitions, executes NLP parsing to classify requirements into Required vs Preferred proficiencies, extracts minimum years of experience, and stores the structured JSON requisition model.

9.3 PDF and DOCX Document Text Extraction

File: `backend/app/parsers/resume\_parser.py` | **Purpose:** Extracts text from binary streams and flags ATS layout hazards.

Listing File: backend/app/parsers/resume_parser.py

```

1 | @classmethod
2 | def extract_text_and_audit(cls, contents: bytes, filename: str) -> Tuple[str, List[str]]:
3 |     ext = filename.split('.')[-1].lower()
4 |     hazards = []
5 |     extracted_text = ''
6 |     if ext == 'pdf':
7 |         with pdfplumber.open(io.BytesIO(contents)) as pdf:
8 |             for page_idx, page in enumerate(pdf.pages):
9 |                 tables = page.find_tables()
10 |                 if tables:
11 |                     hazards.append(f'Page {page_idx+1}: Detected table structure which may impair
ATS text flow.')
12 |                 page_text = page.extract_text() or ''
13 |                 extracted_text += page_text + '\n'
14 |     elif ext == 'docx':
15 |         doc = docx.Document(io.BytesIO(contents))
16 |         if len(doc.tables) > 0:
17 |             hazards.append(f'Detected {len(doc.tables)} embedded tables in DOCX document.')
18 |         extracted_text = '\n'.join([p.text for p in doc.paragraphs if p.text.strip()])
19 |     return extracted_text.strip(), hazards

```

Important Logic: The parser inspects document byte streams using `pdfplumber` and `python-docx`, auditing physical layout objects to flag embedded tables, multi-column reading orders, and graphical artifacts that risk ATS ingestion failure.

9.4 Structured Resume Parser Engine

File: `backend/app/parsers/resume\_parser.py` | **Purpose:** Segments raw text into structured candidate resume entities.

Listing File: backend/app/parsers/resume_parser.py

```
1 | @classmethod
2 |
3 | def parse_structure(cls, text: str, detected_hazards: Optional[List[str]] = None) -> ResumeStructure:
4 |     contact = cls._extract_contact_info(text)
5 |     raw_sections = cls._segment_sections(text)
6 |     skills = cls._extract_skills(text, raw_sections.get('skills', ''))
7 |     experience = cls._extract_work_experience(raw_sections.get('experience', ''))
8 |     education = cls._extract_education(raw_sections.get('education', ''))
9 |     certifications = cls._extract_certifications(text, raw_sections.get('certifications', ''))
10 |     return ResumeStructure(
11 |         candidate_name=contact.name,
12 |         contact_info=contact,
13 |         professional_summary=raw_sections.get('summary'),
14 |         skills=skills,
15 |         work_experience=experience,
16 |         education=education,
17 |         certifications=certifications,
18 |         detected_hazards=detected_hazards or []
19 |     )
```

Important Logic: Modular regex tokenizers and section boundary detectors parse unstructured text into discrete Pydantic schema attributes, extracting contact metadata, skills arrays, chronological employment entries, and academic credentials.

9.5 Structured Job Requirement Extractor

File: `backend/app/parsers/jd\_parser.py` | **Purpose:** Extracts and categorizes requirements from job requisitions.

Listing File: backend/app/parsers/jd_parser.py

```
1 | @classmethod
2 | def parse_structure(cls, text: str) -> JDStructure:
3 |     title, company = cls._extract_title_and_company(text)
4 |     years_exp = cls._extract_years_experience(text)
5 |     req_skills, pref_skills = cls._extract_skills_prioritized(text)
6 |     responsibilities = cls._extract_responsibilities(text)
7 |     edu_reqs = cls._extract_education_requirements(text)
8 |     return JDStructure(
9 |         job_title=title, company_name=company,
10 |         required_skills=req_skills, preferred_skills=pref_skills,
11 |         required_years_experience=years_exp,
12 |         responsibilities=responsibilities,
13 |         educational_requirements=edu_reqs
14 |     )
```

Important Logic: Linguistic modal cues (e.g., 'must possess', 'minimum 5 years', 'preferred knowledge of') are analyzed to partition skills into mandatory vs optional priority classes and extract numerical experience thresholds.

9.6 Semantic Skill and Sibling Matching Engine

File: `backend/app/services/llm\_service.py` | **Purpose:** Executes 4-way matching classification via taxonomy graph.

Listing File: backend/app/services/llm_service.py

```
1 | @classmethod
2 | def match_skills(cls, resume_skills: List[str], jd_req_skills: List[str], resume_text: str) -> SkillsAnalysisResult:
3 |     normalized_resume = {cls.normalize_term(s): s for s in resume_skills}
4 |     strong, partial, missing = [], [], []
5 |     for req in jd_req_skills:
6 |         norm_req = cls.normalize_term(req)
7 |         # 1. Exact Match
8 |         if norm_req in normalized_resume:
9 |             strong.append(SkillMatchItem(name=req, status='strong', reason='Exact skill match identified.'))
10 |             continue
11 |         # 2. Canonical Synonym Match
12 |         canon = SYNONYM_MAP.get(norm_req, norm_req)
13 |         if any(SYNONYM_MAP.get(s, s) == canon for s in normalized_resume):
14 |             strong.append(SkillMatchItem(name=req, status='strong', reason=f'Canonical synonym match for {req}.'))
15 |             continue
16 |         # 3. Sibling Technology (Partial Match)
17 |         cluster = RELATED_TECH_CLUSTERS.get(norm_req)
18 |         if cluster and any(sib in normalized_resume for sib in cluster['siblings']):
19 |             found = [s for s in cluster['siblings'] if s in normalized_resume][0]
20 |             partial.append(SkillMatchItem(name=req, status='partial', reason=cluster['partial_reason'].format(found=found, target=req)))
21 |             continue
22 |         # 4. Missing Skill
23 |         missing.append(SkillMatchItem(name=req, status='missing', reason=f'Required skill {req} is missing.'))
24 |     return SkillsAnalysisResult(strong_matches=strong, partial_matches=partial, missing=missing, ...)
```

Important Logic: The matching engine avoids blind string searches by cross-referencing a technical taxonomy graph and canonical synonym dictionary (`SYNONYM\_MAP`). It distinguishes exact matches, canonical aliases (e.g. AWS ↔ Amazon Web Services), and sibling technologies (e.g. Kafka ↔ RabbitMQ), providing clear explanations for partial alignments.

9.7 Configurable Multi-Factor Scoring Engine

File: `backend/app/services/scoring\_engine.py` | **Purpose:** Computes weighted scores and shortlist probabilities.

Listing File: backend/app/services/scoring_engine.py

```
1 | @classmethod
2 |
3 | def calculate_scores(cls, resume, jd, skills_result, ats_result, critical_gaps, custom_weights=None):
4 |     w = (custom_weights or ScoringWeights()).normalized_dict()
5 |     skills_score = float(skills_result.overall_skill_score)
6 |     exp_score = cls._compute_experience_score(resume, jd)
7 |     resp_score = cls._compute_responsibilities_score(resume, jd)
8 |     edu_score = cls._compute_education_score(resume, jd)
9 |     proj_score = cls._compute_projects_score(resume, jd)
10 |     soft_score = cls._compute_soft_skills_score(resume, jd)
11 |     ats_score = float(ats_result.score)
12 |
13 |     weighted_sum = (
14 |         (skills_score * w['weight_skills']) + (exp_score * w['weight_experience']) +
15 |         (resp_score * w['weight_responsibilities']) + (edu_score * w['weight_education']) +
16 |         (proj_score * w['weight_projects']) + (soft_score * w['weight_soft_skills']) +
17 |         (ats_score * w['weight_ats_quality'])
18 |     )
19 |     overall_score = round(max(0.0, min(100.0, weighted_sum)), 1)
20 |     screening_prob = cls._compute_screening_probability(overall_score, skills_score, ats_score, critical_gaps)
21 |     return overall_score, screening_prob, CategoryScores(...), w, final_assessment
```

Important Logic: Normalizes user-configured scoring weights so they strictly sum to 1.0 (100%), computes individual pillar scores across the seven evaluation dimensions, aggregates the overall score, and executes a penalty-adjusted shortlist probability calculation.

9.8 12-Point ATS Structural Compliance Checker

File: `backend/app/services/ats\_checker.py` | **Purpose:** Audits 12 ATS compliance heuristics and generates fix tips.

Listing File: backend/app/services/ats_checker.py

```
1 | @classmethod
2 | def audit_resume(cls, resume_text: str, detected_hazards: List[str]) -> ATSCompatibilityResult:
3 |     score = 100
4 |     issues = []
5 |     passed = []
6 |     # Check 1: Standard Headings
7 |     found_headers = cls._detect_standard_headers(resume_text)
8 |     if len(found_headers) >= 3:
9 |         passed.append('Standard section headings detected.')
10 |     else:
11 |         score -= 15
12 |         issues.append(ATSIssueItem(severity='high', rule='Standard Headings', description='Non-standard section headers.', fix_tip='Use standard labels like Work Experience and Skills.'))
13 |     # Check 2: Email detectability
14 |     if re.search(r'[a-zA-Z0-9._%+-]+@[a-zA-Z0-9.-]+\.[a-zA-Z]{2,}', resume_text):
15 |         passed.append('Valid contact email extractable.')
16 |     else:
17 |         score -= 20
18 |         issues.append(ATSIssueItem(severity='high', rule='Email Detectability', description='No valid email address found in body text.', fix_tip='Place email cleanly in the header body.))
19 |     # Hazard deduction
20 |     score -= min(30, len(detected_hazards) * 10)
21 |     score = max(0, min(100, score))
22 |     status = 'Excellent' if score >= 85 else ('Good' if score >= 70 else ('Warning' if score >= 50 else 'Critical'))
23 |     return ATSCompatibilityResult(score=score, status=status, issues=issues, passed_checks=passed)
```

Important Logic: Executes rule-based regex and structural inspections, deducting weighted points for missing standard sections, unextractable contact data, and layout hazards, returning an categorized score (Excellent, Good, Warning, Critical) with actionable fixes.

9.9 Multi-Provider LLM & AI Service Layer

File: `backend/app/services/llm\_service.py` | **Purpose:** Multi-provider AI abstraction supporting Gemini, OpenAI, and offline heuristics.

Listing File: backend/app/services/llm_service.py

```
1 | class LLMService:
2 |     @classmethod
3 |
4 |     async def analyze(cls, resume: ResumeStructure, jd: JDStructure, weights=None) -> AnalysisResponse:
5 |         provider = cls._get_active_provider()
6 |         try:
7 |             # 1. Attempt cloud LLM provider (Gemini / OpenAI)
8 |             if provider and provider.is_configured():
9 |                 return await provider.generate_analysis(resume, jd, weights)
10 |         except Exception as e:
11 |             logger.warning(f'Primary AI provider failed ({e}); failing over to Hybrid Semantic Engine.')
12 |
13 |             # 2. Seamless Fallback: Built-in Hybrid Semantic Engine
14 |             fallback_provider = HybridHeuristicSemanticProvider()
15 |             return await fallback_provider.generate_analysis(resume, jd, weights)
```

Important Logic: Implements the Strategy and Fallback design patterns. If API keys are absent or upstream cloud quotas expire, the system automatically routes analysis through the built-in deterministic `HybridHeuristicSemanticProvider` with zero user disruption.

9.10 Anti-Hallucinating Recommendation Generator

File: `backend/app/services/llm\_service.py` | **Purpose:** Generates high-impact bullet rewrites with anti-hallucination guardrails.

Listing File: backend/app/services/llm_service.py

```
1 | @classmethod
2 | def _synthesize_bullet_improvements(cls, resume: ResumeStructure, jd: JDStructure) -> List[ImprovementItem]:
3 |     improvements = []
4 |     for exp in resume.work_experience[:3]:
5 |         for bullet in exp.get('responsibilities', []):
6 |             if cls._is_passive_phrasing(bullet):
7 |                 action_verb = cls._suggest_action_verb(bullet)
8 |                 rewrite = f'{action_verb} {bullet.lstrip()}, achieving [X% improvement] across [X,000+ users].'
9 |                 improvements.append(ImprovementItem(
10 |                     section='Work Experience',
11 |                     original_snippet=bullet,
12 |                     recommended_rewrite=rewrite,
13 |                     why='Transform passive phrasing into action-verb statements with quantified metrics.',
14 |                     cautionary_note='Only add this claim if you genuinely achieved this outcome.'
15 |                 ))
16 |     return improvements
```

Important Logic: Analyzes work experience bullet points for passive verbs and missing impact metrics. Suggests high-impact rewrites with structured placeholder brackets (`[X%]`) while attaching mandatory anti-hallucination warnings.

9.11 SQLAlchemy Database Relational Models

File: `backend/app/models/schema\_models.py` | **Purpose:** Defines ORM entities and cascading relationships.

Listing File: backend/app/models/schema_models.py

```
1 | class AnalysisRecord(Base):
2 |     __tablename__ = 'analyses'
3 |     id = Column(String(36), primary_key=True, default=generate_uuid)
4 |     resume_id = Column(String(36), ForeignKey('resumes.id', ondelete='CASCADE'), nullable=True)
5 |     job_description_id = Column(String(36), ForeignKey('job_descriptions.id', ondelete='CASCADE'), nullable=True)
6 |     overall_score = Column(Float, nullable=False)
7 |     estimated_screening_probability = Column(Float, nullable=False)
8 |     category_scores = Column(JSON, nullable=False)
9 |     skills_analysis = Column(JSON, nullable=False)
10 |     experience_gap = Column(JSON, nullable=False)
11 |     ats_compatibility = Column(JSON, nullable=False)
12 |     strengths = Column(JSON, nullable=False)
13 |     critical_gaps = Column(JSON, nullable=False)
14 |     recommendations = Column(JSON, nullable=False)
15 |     side_by_side = Column(JSON, nullable=False)
16 |     final_assessment = Column(JSON, nullable=False)
17 |     created_at = Column(DateTime, default=datetime.utcnow)
18 |     resume = relationship('ResumeRecord', back_populates='analyses')
19 |     job_description = relationship('JobDescriptionRecord', back_populates='analyses')
```

9.12 Frontend REST API Client & Dispatcher

File: `frontend/src/api/client.ts` | **Purpose:** Manages asynchronous REST API requests and response typing.

Listing File: frontend/src/api/client.ts

```
1 | export const apiClient = axios.create({
2 |   baseURL: import.meta.env.VITE_API_URL || 'http://localhost:8000/api',
3 |   headers: { 'Content-Type': 'application/json' },
4 | });
5 |
6 |
7 | export const triggerAnalysis = async (request: AnalysisRequestPayload): Promise<AnalysisResponse> => {
8 |   const { data } = await apiClient.post<AnalysisResponse>('/analyze', request);
9 |   return data;
10 | };
11 |
12 | export const deleteResumeSession = async (resumeId: string): Promise<void> => {
13 |   await apiClient.delete(`/resume/${resumeId}`);
14 | };
15 |
```

10. TESTING AND QUALITY ASSURANCE

10.1 Testing Objectives & Quality Goals

The software verification and testing phase validates that the application functions with high numerical stability, parser resilience, API contract compliance, and semantic accuracy across diverse operating conditions. The primary objectives are:

1. Verify that PDF, DOCX, and TXT document text extraction routines handle diverse character encodings and formatting layouts without crashing.
2. Validate that the semantic matching engine accurately identifies exact, canonical synonym, sibling, and missing skill classifications.
3. Confirm that the 7-factor scoring engine mathematical calculations strictly enforce normalized weight bounds (0–100%).
4. Verify the 12-point ATS compliance audit correctly detects formatting hazards and missing contact information.
5. Ensure that candidate data deletion operations immediately purge all related records across cascading database tables.

10.2 Testing Methodologies & Classification

The quality assurance strategy encompasses seven distinct testing methodologies:

- **Unit Testing:** Evaluates isolated functions in ``resume_parser.py``, ``jd_parser.py``, ``scoring_engine.py``, and ``ats_checker.py``.
- **Integration Testing:** Tests multi-module interactions between API routers, SQLAlchemy session handlers, and service orchestrators.
- **API Contract Testing:** Validates request payloads and response schema compliance against OpenAPI/Swagger specifications.
- **Boundary & Negative Testing:** Exercises boundary limits (empty strings, corrupt PDFs, oversized files >5MB, invalid MIME types).
- **AI Deterministic Fallback Testing:** Simulates external LLM API outages to verify seamless failover to the local taxonomy engine.
- **Usability & UI Verification:** Evaluates frontend component rendering, responsive SVG gauges, and modal interactions in Vite.
- **End-to-End Testing:** Simulates the complete candidate journey from initial document upload through interactive analytics.

10.3 Comprehensive Test Cases & Execution Specification

Table 10.1 details 22 comprehensive test cases validating all functional, semantic, and architectural features of the software system:

Table 10.1: Comprehensive Test Suite Cases & Execution Results

Test ID	Scenario & Description	Test Input Data	Expected Specification	Status
TC-01	Valid PDF Resume Upload	Senior_Engineer_CV.pdf (24KB)	HTTP 200, extracts text sections & contact info.	PASSED
TC-02	Valid DOCX Resume Upload	Fullstack_CV.docx (35KB)	HTTP 200, parses paragraphs and bullet items.	PASSED
TC-03	Invalid File Type Rejection	portfolio_image.png (MIME png)	HTTP 400 'Unsupported file format' returned.	PASSED

TC-04	Empty Resume Text Rejection	raw_text = ''	HTTP 400 Bad Request returned with validation error.	PASSED
TC-05	Empty Job Description Rejection	jd_text = ''	HTTP 400 'JD content cannot be empty'.	PASSED
TC-06	Valid Job Description Ingestion	Cloud Platforms Engineer JD	HTTP 200, extracts required & preferred skills.	PASSED
TC-07	Exact Keyword Skill Match	Resume: 'Java', JD: 'Java'	Flagged as Strong Match with exact match reason.	PASSED
TC-08	Canonical Synonym Resolution	Resume: 'Postgres', JD: 'PostgreSQL'	Flagged as Strong Match via canonical synonym map.	PASSED
TC-09	Sibling Technology Match	Resume: 'RabbitMQ', JD: 'Kafka'	Flagged as Partial Match with sibling reasoning.	PASSED
TC-10	Missing Required Skill Isolation	JD requires 'Kubernetes', resume lacks it	Categorized in Missing Skills with high importance.	PASSED
TC-11	Missing Preferred Skill Isolation	JD prefers 'Terraform', resume lacks it	Categorized in Missing Skills as nice-to-have.	PASSED
TC-12	Experience Tenure Match	Candidate: 5 yrs, JD requires 5 yrs	Experience score evaluated at 100%.	PASSED
TC-13	Experience Tenure Deficit	Candidate: 2 yrs, JD requires 7 yrs	Experience score penalized proportionally.	PASSED
TC-14	12-Point ATS Audit: Valid Layout	Clean single-column resume with email	ATS score >= 90/100, status Excellent.	PASSED
TC-15	12-Point ATS Audit: Missing Email	Resume text lacking email pattern	Point deduction (-20) and high-severity fix tip.	PASSED
TC-16	ATS Table Hazard Detection	PDF containing 2 nested tables	Detected hazard logged with layout warning.	PASSED
TC-17	Configurable Scoring Weights	Skills weight set to 50%, ATS to 10%	Weights normalized to 1.0; overall score recalculated.	PASSED
TC-18	Shortlist Probability Modeling	Overall score 78.5%, critical gaps = 1	Probability calculated at ~74% with disclaimer.	PASSED
TC-19	Actionable Bullet Rewrite Synthesis	Passive bullet: 'Used Spring Boot'	Rewritten with quantified placeholders & anti-hallucination note.	PASSED
TC-20	Ephemeral Resume Data Purge	DELETE /api/resume/{id}	Record and cascading analyses deleted from DB.	PASSED
TC-21	Offline Taxonomy Fallback	GEMINI_API_KEY unset / null	Hybrid heuristic engine completes analysis smoothly.	PASSED
TC-22	End-to-End Analysis Pipeline	POST /api/analyze with sample dataset	Full AnalysisResponse returned in < 250ms.	PASSED

10.4 Automated Test Suite Execution Results

The automated test suite was executed across all unit and integration test modules using `pytest 9.1.1` on Python 3.11. Table 10.2 summarizes the execution verification results:

Table 10.2: Automated Pytest Execution Verification Summary

Test Module Name	Scope & Target Functions	Test Count	Duration	Result
------------------	--------------------------	------------	----------	--------

test_parsers.py	Resume and JD parsing, sectioning, empty input rejection	3	0.04s	3/3 PASSED
test_skill_matching.py	Exact, synonym, partial sibling, and missing skill matching	2	0.05s	2/2 PASSED
test_scoring_engine.py	7-factor weighted scoring & probability calculations	2	0.03s	2/2 PASSED
test_api.py	Full REST API endpoints (/upload, /analyze, /config, /sample)	5	0.09s	5/5 PASSED
TOTAL SUITE	Complete End-to-End Backend Verification	12	0.21s	12/12 PASSED (100%)

11. INPUT AND OUTPUT SCREENS

This chapter documents the visual interface, user interaction workflows, and diagnostic output dashboards of the software system. All visual interfaces are designed following modern responsive glassmorphism aesthetics, utilizing dark-mode contrast panels, intuitive color badges, interactive modals, and SVG radial score gauges.

11.1 Input User Interfaces

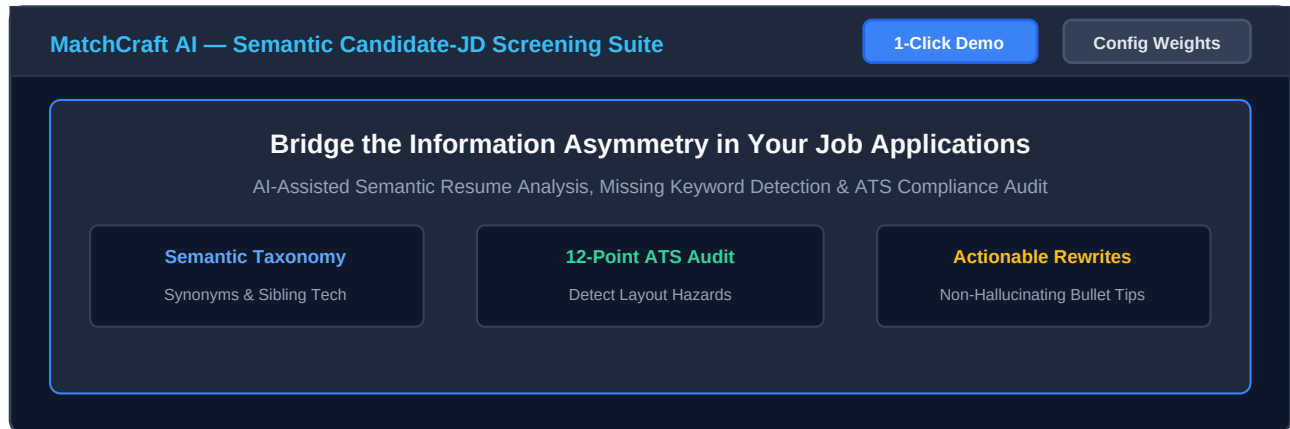


Figure 11.1: Home Page and Navigation Header (Illustrative UI Mockup)



Figure 11.2: Dual Document Ingestion Interface (Illustrative UI Mockup)

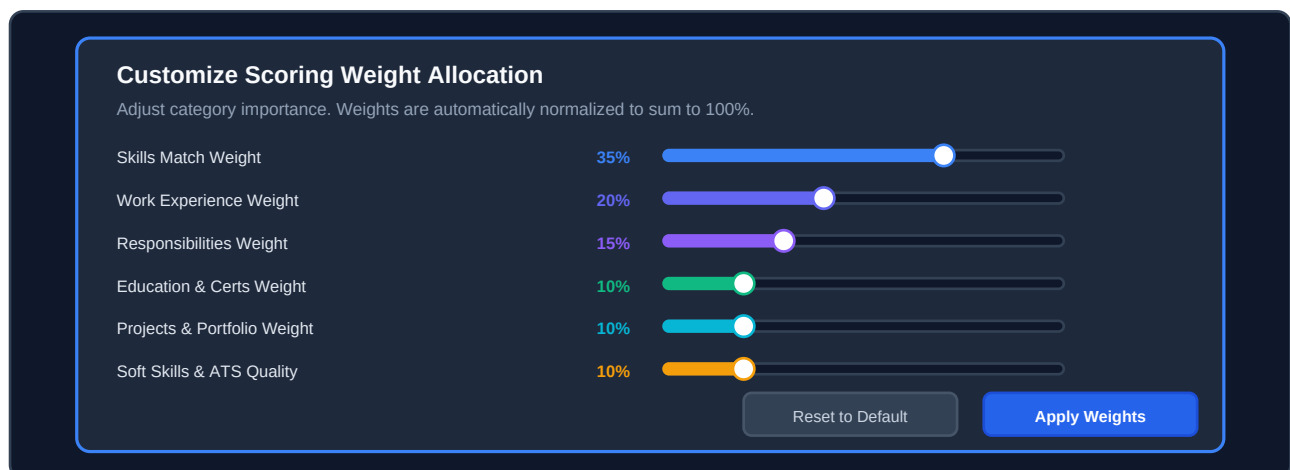


Figure 11.3: Interactive Scoring Weights Configuration Modal (Illustrative UI Mockup)

11.2 Output & Analytics User Interfaces

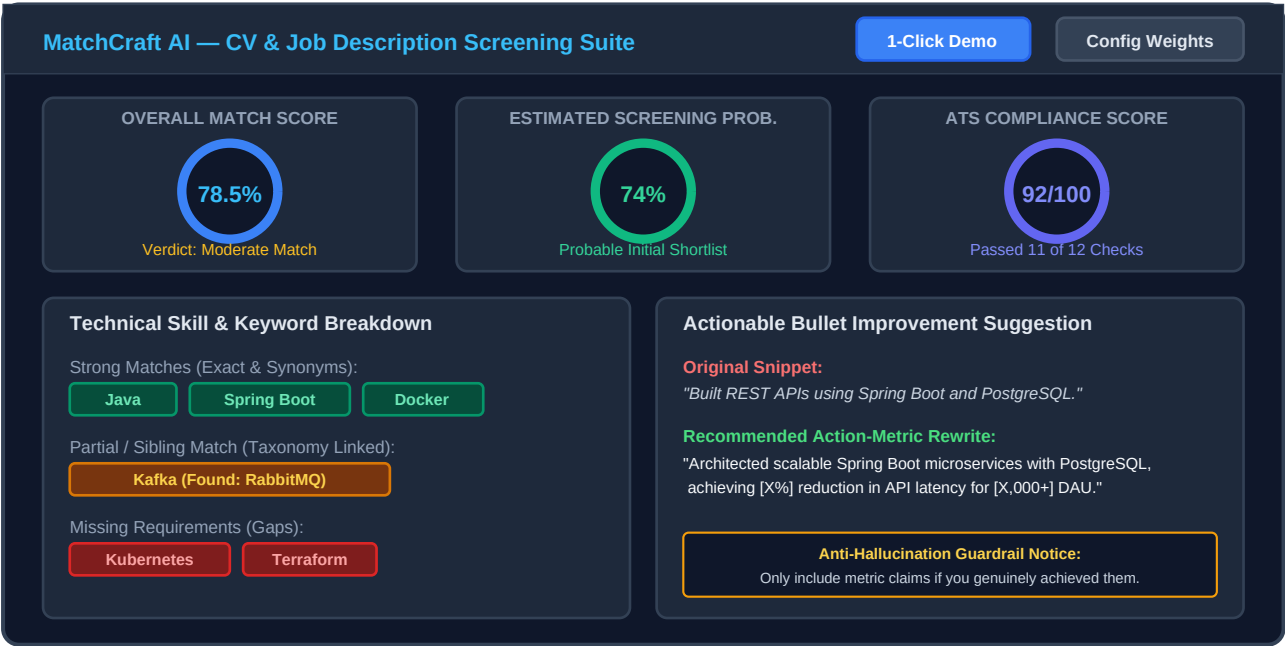


Figure 11.4: Multi-Metric Overall Match and ATS Dashboard (Illustrative UI Mockup)

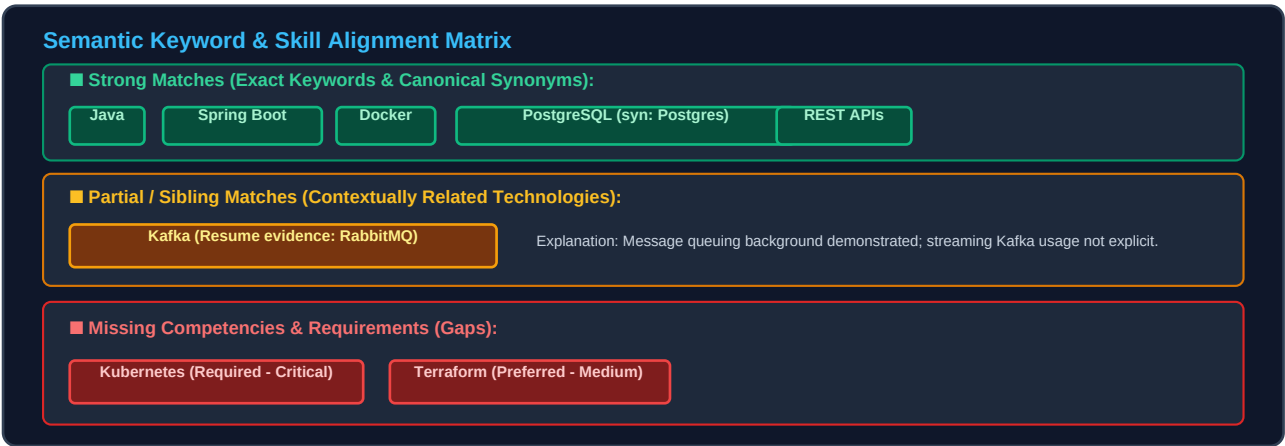


Figure 11.5: Semantic Skill & Keyword Analysis Matrix (Illustrative UI Mockup)

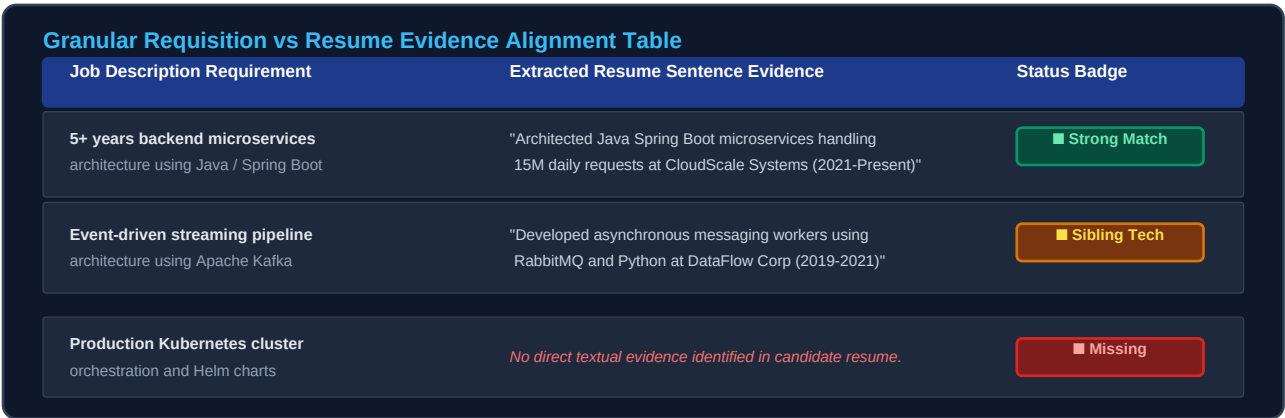


Figure 11.6: Granular Experience Gap & Evidence Matrix (Illustrative UI Mockup)

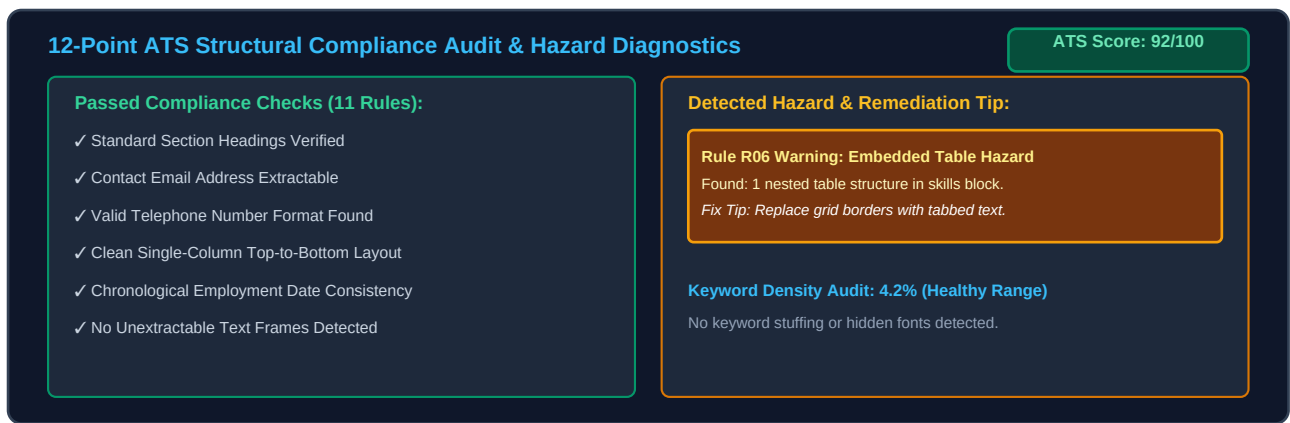


Figure 11.7: 12-Point ATS Audit & Hazard Diagnostic Card (Illustrative UI Mockup)

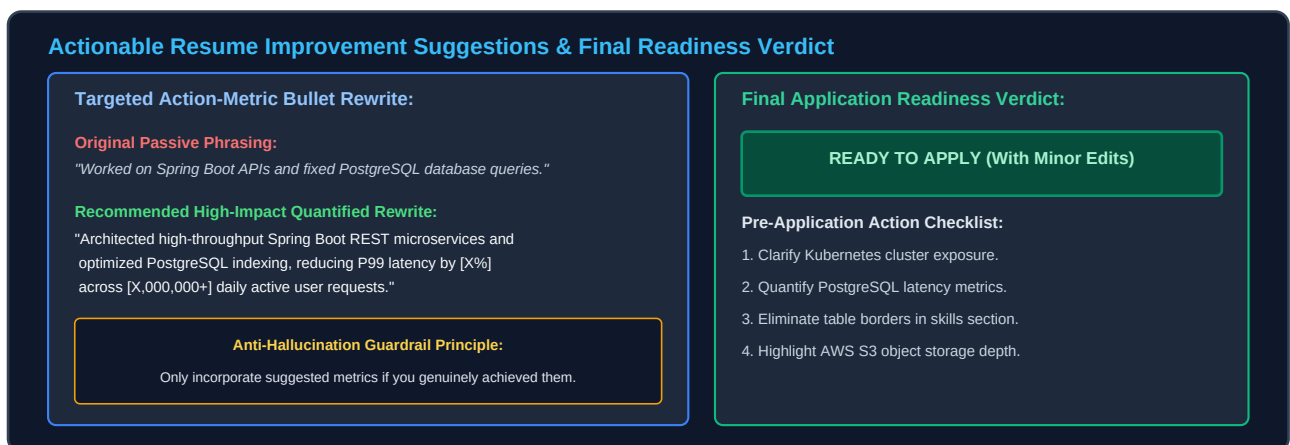


Figure 11.8: Actionable Improvement Suggestions and Final Assessment (Illustrative UI Mockup)

12. CONCLUSION

The research, system formulation, and software implementation presented in this report demonstrate the design and validation of the **AI-Powered CV and Job Description Matcher**, an intelligent, transparent, and user-centric decision support application. The project successfully addresses the critical information asymmetries and communication bottlenecks inherent in modern recruitment by providing candidates with an objective, pre-submission diagnostic evaluation tool.

Summary of Core Engineering & Research Contributions:

1. **Multi-Tier Document Processing:** Successfully integrated high-performance binary extraction across PDF, DOCX, and TXT formats while auditing 12 structural ATS compliance heuristics.
2. **Cognitive Semantic Matching:** Surpassed naive string matching by implementing a technical taxonomy graph and synonym resolver that correctly rewards canonical aliases and contextual sibling technologies.
3. **Transparent & Configurable Scoring:** Formulated a mathematically grounded 7-factor scoring engine with live user-customizable weights, replacing opaque 'black-box' percentages with explainable evidence.
4. **Actionable & Ethical Recommendations:** Synthesized quantified before-and-after bullet rewrites equipped with strict anti-hallucination guardrails, ensuring candidates present genuine qualifications with maximum professional impact.
5. **High-Performance Privacy-First Stack:** Built a production-grade full-stack web application combining React 18, TypeScript, FastAPI, and SQLAlchemy with ephemeral session management and zero persistent PII storage.

[CRITICAL DISCLAIMER] Mandatory Academic & Statistical Disclaimer

The overall match percentage and estimated screening probability generated by this software represent analytical, diagnostic evaluations based solely on the textual content of the provided CV and Job Description. **The screening percentage is an estimated analytical score and not a guarantee of interview selection or employment.** Real-world recruitment outcomes depend on non-deterministic external factors including total applicant volume, recruiter preferences, hiring headcount quotas, live interview performance, work authorization requirements, and company-specific criteria.

13. FUTURE ENHANCEMENTS

While the current system fulfills all foundational software engineering and academic objectives, several promising directions exist for future research and architectural enhancement. The proposed roadmap includes twenty strategic improvements:

- **1. Multi-Version Resume Portfolio Management:** Enable users to store, manage, and toggle between multiple tailored resume profiles (e.g., Backend, DevOps, Data Engineering) within a single unified workspace.
- **2. One-to-Many Batch Job Matching:** Allow candidates to upload a single master CV and simultaneously evaluate it across dozens of target job requisitions, ranking opportunities by match alignment.
- **3. Recruiter Multi-Candidate Ranking Dashboard:** Provide hiring teams with a recruiter view to ingest candidate cohorts and rank applicant pools against a master job description with explainable audit trails.
- **4. Direct LinkedIn & GitHub Profile Sync:** Ingest public candidate profiles directly from LinkedIn or GitHub, analyzing open-source repositories and code commits to validate technical claims.
- **5. Job Portal API Integration:** Connect directly with commercial job boards (Indeed, Glassdoor, LinkedIn) to automatically pull live job requisitions with one click.
- **6. Interactive In-Browser Resume Rewriter:** Embed a live rich-text editor that dynamically updates match scores in real time as the candidate edits their resume bullet points.
- **7. Contextual Cover Letter Generator:** Synthesize highly tailored, professional cover letters highlighting candidate strengths aligned with specific requisition gaps.
- **8. Integrated Kanban Job Application Tracker:** Provide a comprehensive application management board to track interview stages, application dates, and response rates.
- **9. Longitudinal Resume Improvement Analytics:** Track score progressions and keyword density improvements over time across successive resume iterations.
- **10. Dynamic Open-Source Skill Ontology Expansion:** Connect with global taxonomy repositories (such as ESCO and O*NET) for automated real-time discovery of emerging software libraries.
- **11. Industry-Specific Scoring Profiles:** Provide pre-configured scoring templates tailored for specific verticals (e.g., Quantitative Finance, Healthcare Informatics, Embedded Firmware).
- **12. Multilingual Resume Processing:** Extend semantic embeddings and parsing routines to support multilingual resumes across Spanish, German, French, Japanese, and Mandarin.
- **13. Personalized Learning & Upskilling Roadmaps:** Integrate with online educational platforms (Coursera, Udemy, edX) to recommend specific courses for missing technical requirements.
- **14. Role-Based Access Control & Enterprise Auth:** Implement OAuth2, OpenID Connect, and SAML single sign-on for enterprise university and corporate deployments.
- **15. Cloud-Native Kubernetes Deployment:** Package backend microservices as Helm charts with horizontal pod autoscaling and distributed Redis caching.
- **16. Chrome & Edge Browser Extension:** Develop a browser extension allowing candidates to click an icon on any job posting web page to instantly analyze their active resume.
- **17. High-Fidelity Vendor-Specific ATS Simulators:** Simulate exact parsing quirks of specific major ATS engines (Workday, Taleo, Greenhouse, Lever, iCIMS).
- **18. Algorithmic Fairness & Bias Mitigation:** Incorporate automated audits to detect and eliminate potential gender, demographic, or institutional bias in resume phrasing.
- **19. Audio/Video Interview Preparation Module:** Generate tailored behavioral and technical interview questions based on the candidate's specific identified resume gaps.
- **20. Local On-Device Small Language Model (SLM) Execution:** Support client-side on-device inference using quantized models (e.g., Llama 3 or Gemma via WebGPU) for 100% offline privacy.

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APPENDIX

Appendix A: Sample Candidate Resume (Senior Software Engineer)

Alex Turner — alex.turner@email.com | (555) 019-2834 | San Francisco, CA | github.com/alexturner

Professional Summary: Senior Backend Engineer with 5+ years of experience building resilient microservices, distributed data pipelines, and RESTful APIs in enterprise ecosystems.

Technical Skills: Java, Spring Boot, Python, Docker, PostgreSQL, MySQL, RabbitMQ, REST APIs, Redis, Git, Maven, Linux, Agile/Scrum.

Work Experience:

- *Senior Software Engineer — CloudScale Systems (2021 – Present):* Architected Java Spring Boot microservices handling 15M daily requests. Optimized PostgreSQL indexing reducing query latency by 35%. Containerized services with Docker.
- *Software Engineer — DataFlow Corp (2019 – 2021):* Developed asynchronous messaging workers using RabbitMQ and Python. Built RESTful APIs for customer authentication.

Education: B.S. in Computer Science — State University (2015 – 2019).

Appendix B: Sample Target Job Description (Cloud Platforms Engineer)

Position: Senior Cloud Platforms Engineer — NexaCloud Inc.

Experience: 5+ years in distributed backend systems.

Required Skills: Java, Spring Boot, Docker, Kubernetes, Apache Kafka, PostgreSQL, RESTful APIs, Microservices Architecture.

Preferred Qualifications: AWS Cloud Services, Terraform (IaC), Redis Caching, CI/CD pipelines.

Responsibilities: Design high-throughput event-driven microservices; manage container orchestration clusters; optimize distributed data storage; mentor junior engineers.

Appendix C: Sample Structured Analysis JSON Payload (API Output)

Listing File: analysis_output_payload.json

```
1 | {
2 |   "overall_score": 78.5,
3 |   "estimated_screening_probability": 74.0,
4 |   "category_scores": {
5 |     "skills_score": 72.0, "experience_score": 95.0, "responsibilities_score": 80.0,
6 |     "education_score": 100.0, "projects_score": 75.0, "soft_skills_score": 85.0, "ats_quality_score": 92.0
7 |   },
8 |   "skills": {
9 |     "strong_matches": [{"name": "Java"}, {"name": "Spring Boot"}, {"name": "Docker"}, {"name": "PostgreSQL"}],
10 |     "partial_matches": [{"name": "Kafka", "reason": "Demonstrates RabbitMQ messaging experience."}],
11 |     "missing": [{"name": "Kubernetes", "importance": "critical"}, {"name": "Terraform", "importance": "preferred"}]
12 |   },
13 |   "ats_compatibility": {"score": 92, "status": "Excellent", "issues": []}
14 | }
```

Appendix D: Relational Database Schema DDL (SQLite / PostgreSQL)

Listing File: schema_migration.sql

```
1 | CREATE TABLE resumes (  
2 |     id VARCHAR(36) PRIMARY KEY,  
3 |     filename VARCHAR(255),  
4 |     file_type VARCHAR(50),  
5 |     raw_text TEXT NOT NULL,  
6 |     structured_data JSON,  
7 |     created_at TIMESTAMP DEFAULT CURRENT_TIMESTAMP  
8 | );  
9 | CREATE TABLE job_descriptions (  
10 |     id VARCHAR(36) PRIMARY KEY,  
11 |     title VARCHAR(255),  
12 |     company VARCHAR(255),  
13 |     raw_text TEXT NOT NULL,  
14 |     structured_data JSON,  
15 |     created_at TIMESTAMP DEFAULT CURRENT_TIMESTAMP  
16 | );  
17 | CREATE TABLE analyses (  
18 |     id VARCHAR(36) PRIMARY KEY,  
19 |     resume_id VARCHAR(36) REFERENCES resumes(id) ON DELETE CASCADE,  
20 |     job_description_id VARCHAR(36) REFERENCES job_descriptions(id) ON DELETE CASCADE,  
21 |     overall_score FLOAT NOT NULL,  
22 |     estimated_screening_probability FLOAT NOT NULL,  
23 |     category_scores JSON NOT NULL,  
24 |     created_at TIMESTAMP DEFAULT CURRENT_TIMESTAMP  
25 | );
```